

Article

Reserve Deliverability-Constrained Battery Energy Storage Sizing for Renewable Power Plants Considering Cycling Degradation

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Abstract

High penetration of wind and photovoltaic generation requires renewable power plants to maintain not only energy delivery capability, but also prescribed upward and downward reserve capability. However, many existing battery energy storage system (BESS) sizing models evaluate reserve capability mainly through nominal power headroom or optional ancillary service revenue, which may lead to biased capacity decisions when state-of-charge limits and point of common coupling (PCC) constraints are binding. This paper proposes a reserve deliverability-constrained BESS sizing framework for renewable power plants considering cycling degradation. The proposed model represents reserve requirements explicitly and evaluates deliverable reserve by jointly considering BESS power headroom, dual-endpoint state-of-charge energy margins, and PCC export/import headroom. Real day-mapped typical scenarios and extreme stress test scenarios are integrated into the evaluation, while rainflow degradation is assessed after dispatch and included only in the candidate-level economic comparison. Numerical experiments are conducted on an equivalent large-scale renewable portfolio constructed from the 2023 nationally aggregated Belgian wind and photovoltaic generation profiles. Within the original 15-candidate set, the selected configuration increases from 300 MW/300 MWh to 1000 MW/1000 MWh when the reserve requirement rises from 5% to 15%. Under the base economic assumptions and the cycling-only degradation model, tightening the PCC export limit from 10000 to 4000 MW changes the best-performing configuration from 300 MW/300 MWh/1 h to 1000 MW/4000 MWh/4 h. Ablation results indicate that ignoring reserve requirements or deliverability constraints may lead to smaller selected capacities or overestimated feasible reserve capability. The proposed framework provides an interpretable sizing method for renewable plant-level BESS planning under prescribed reserve requirements, PCC constraints, and cycling degradation costs.

Keywords: battery energy storage system; renewable power plant; reserve deliverability; energy storage sizing



Academic Editors: Xionghu Zhong and Jielong Yang

Received: 6 July 2026

Revised: 27 July 2026

Accepted: 28 July 2026

Published: 3 August 2026

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1. Introduction

The rapid growth of wind and photovoltaic (PV) generation has changed the operating requirements of modern power systems. Compared with conventional synchronous units,

renewable energy resources have stronger output uncertainty, weaker controllability, and limited inherent inertia. These characteristics increase the need for flexible resources that can support energy balancing, operational flexibility, and system security. Battery energy storage systems (BESSs) are widely considered in this context because of their fast response, modular deployment, and ability to participate in multiple grid services. Recent international reports have also emphasized the role of large-scale battery deployment in renewable energy integration and clean energy transitions [1,2]. Therefore, appropriate BESS sizing for renewable power plants has become an important planning problem.

Existing studies have formulated BESS planning problems under various operational and economic objectives, including energy arbitrage, renewable curtailment reduction, peak shaving, reliability improvement, and multi-service revenue stacking. Reviews have summarized BESS sizing and siting methods, application scenarios, degradation models, objective functions, and operational constraints [3,4]. In renewable-rich systems, storage is often planned to absorb surplus generation and release it during later periods, thereby improving renewable energy utilization [5]. These studies provide useful foundations for renewable storage planning. However, many models still evaluate BESS value mainly from an energy-shifting or market revenue perspective. In such formulations, reserve provision is often treated as an optional ancillary service revenue source rather than as a prescribed capability that should be satisfied during capacity planning. For renewable power plants required to maintain upward and downward reserve capability, this treatment may lead to insufficient BESS capacity.

Reserve capability should also be evaluated from the perspective of deliverability. A common simplification is to estimate available reserve from the remaining BESS power margin, such as the difference between the rated power and scheduled charging or discharging power. This approximation may be acceptable for a rough estimate, but it can overstate the reserve that is actually deliverable at the plant-grid interface. To provide upward or downward reserve, the BESS must retain a sufficient state of charge (SOC) margin to sustain the reserved power for the required deployment duration. This margin should remain feasible both before and after the scheduled charging or discharging action within the same interval. In addition, reserve delivery is limited by the point of common coupling (PCC). When the baseline power exchange already uses a large portion of the PCC export or import capacity, additional reserve may not be deliverable even if the BESS itself still has apparent power headroom. Although previous studies have discussed converter capability, security limits, and device-level feasibility in BESS ancillary service control [6], their combined impact with SOC energy margins and PCC headroom has not been sufficiently reflected in renewable plant-level BESS sizing models.

Battery degradation and scenario representation are also important in BESS sizing. The life-cycle economic performance of a storage system is affected not only by operating revenue, but also by cycle aging caused by repeated charging and discharging. Such aging is related to the SOC trajectory, depth of discharge, equivalent full cycles, and service-specific operating patterns. Recent studies have developed degradation-aware scheduling models and degradation cost formulations, including rainflow-based approaches [7,8]. However, directly embedding detailed rainflow counting into an MILP dispatch model can introduce nonlinear temporal coupling and increase computational complexity. A practical planning method therefore needs to account for degradation while keeping the dispatch model tractable [9]. At the same time, the chronological shape of renewable generation affects storage operation and SOC cycling. Representative-day methods are commonly used to reduce computational burden, but clustering centroids may smooth short-term ramps and peak-valley variations, leading to biased SOC and degradation

estimates. This motivates the use of scenario construction methods that preserve realistic intra-day fluctuations without greatly increasing the number of dispatch problems [10].

Motivated by these issues, this paper develops a reserve deliverability-constrained BESS sizing framework for renewable power plants considering cycling degradation. The framework models upward and downward reserve requirements as explicit planning constraints, rather than treating reserve provision only as an optional source of ancillary service revenue. Reserve deliverability is evaluated through constraints that combine BESS power headroom, dual-endpoint SOC energy margins, and PCC export/import headroom. In addition, real day-mapped typical scenarios, extreme stress test scenarios, and rainflow-based degradation post-processing are integrated into the capacity evaluation process so that candidate BESS configurations can be compared under both economic and operational considerations.

The main contributions of this paper are as follows:

1. A renewable plant-level BESS sizing framework is developed by modeling prescribed upward and downward reserve requirements as explicit planning constraints. Different from energy-only or revenue-stacking formulations, the proposed model can examine how increasing reserve requirements affect the selected BESS configuration within a predefined candidate set.
2. A reserve deliverability constraint set is formulated by jointly considering BESS power headroom, dual-endpoint SOC energy margins, and PCC export/import headroom. This formulation reduces the risk of overestimating reserve capability based solely on the rated power or instantaneous power headroom of the BESS.
3. A scenario-based sizing and evaluation workflow is established by combining real day-mapped typical scenarios, extreme stress test scenarios, and rainflow-based degradation post-processing. Compared with directly using clustering centroids, the real-day mapping strategy preserves intra-day renewable fluctuation patterns that are relevant to SOC cycling and degradation evaluation.

The rest of this paper is organized as follows: Section 2 reviews studies on BESS sizing, ancillary service participation, reserve deliverability, degradation modeling, and scenario representation. Section 3 describes the proposed sizing framework, including scenario construction, operational dispatch, reserve deliverability constraints, rainflow degradation assessment, and upper-level capacity evaluation. Section 4 presents the numerical results, including the results of the baseline capacity search, scenario construction comparison, ablation analysis, sensitivity analysis, degradation assessment, and stress test evaluation. Section 5 concludes the paper and outlines future work.

2. Related Work

This section reviews the literature most relevant to the proposed renewable plant-level BESS sizing framework with emphasis on three research streams: BESS sizing for renewable integration, ancillary service participation and reserve deliverability, and degradation-aware scenario representation in storage planning.

2.1. BESS Sizing and Renewable Integration

Battery energy storage systems are now commonly considered in planning studies for renewable-rich power systems because they can provide fast and flexible power adjustment. The associated sizing problem is generally concerned with selecting the rated power and energy capacity of the storage system under specific operating and economic requirements. Typical objectives include reducing renewable curtailment, exploiting energy price differences, shaving peaks, improving reliability, and balancing investment cost against operational benefits. Existing review articles have examined this topic from several

angles considering sizing and siting methods, market participation, policy conditions, environmental effects, and degradation modeling [3,4,11].

For renewable integration, storage is often planned to absorb surplus wind or PV generation and release it in later periods. Wang et al. developed a tri-level siting and sizing model in which energy storage is used to improve renewable generation utilization under multi-stage dispatch uncertainty [5]. Related studies have considered BESS sizing in distribution networks, wind farms, PV–battery systems, and energy communities. In these applications, the benefit of storage is usually linked to curtailed energy recovery, wholesale market arbitrage, investment cost trade-offs, or reliability improvement [12–14]. The representation of time series has also received increasing attention. Studies on temporal resolution and representative period selection indicate that storage planning results can change with the chronological structure of renewable generation profiles and with the way representative periods are selected [10,15–17].

These studies mainly characterize BESSs as energy-shifting resources. From this modeling perspective, the selected capacity is largely shaped by price spreads, curtailment recovery, and investment cost. This treatment is appropriate for many energy-oriented planning problems, but it does not fully describe cases in which a renewable power plant is required to maintain upward and downward reserve capability. In such cases, the sizing result should be evaluated not only by energy value, but also by whether the plant can satisfy prescribed reserve requirements under operational constraints. This paper therefore adopts a reserve requirement-oriented sizing perspective for renewable plant-level BESS planning.

2.2. Ancillary Service Participation and Reserve Deliverability

Another important research stream is investigating BESS participation in ancillary services. Owing to their fast response capability, BESSs are suitable for frequency regulation, voltage support, black starts, congestion relief, peak shaving, and power smoothing [18,19]. More recent studies and reviews have further discussed the role of BESSs in frequency control ancillary services, fast frequency response, renewable energy communities, and multi-market participation [20–23]. Market-oriented studies have also shown that storage participation in energy, reserve, and regulation markets is affected by market rules, degradation costs, bid commitments, and technical requirements [24–27].

Market-based studies have clarified how BESSs can obtain additional value from multi-service operation. However, this type of revenue-stacking formulation is different from a sizing problem in which reserve capability is treated as a planning requirement. In many market-oriented models, ancillary service provision is optional and is scheduled when it improves the operating objective. For a renewable power plant that is required to maintain a prescribed upward and downward reserve capability, the reserve requirement needs to be represented directly in the capacity evaluation; otherwise, the selected BESS configuration may be economically attractive under revenue stacking but still insufficient for meeting the required reserve capability across the considered operating conditions.

Reserve evaluation also depends on whether the scheduled reserve can be delivered at the plant–grid interface. A common simplification is to estimate available reserve from the remaining rated power of the BESS. This treatment is convenient, but it may give an optimistic estimate when other physical limits are binding. Even with nominal power headroom, reserve delivery may be restricted by the SOC boundary, the required deployment duration, or the available export/import margin at the PCC. For example, if the baseline energy schedule already uses most of the PCC capacity, additional upward or downward reserve may not be deliverable to the grid. Operational studies have shown that ancillary service provision should satisfy converter capability, security constraints,

voltage-dependent limits, and device-level feasibility [6,28]. In renewable plant-level sizing models, however, BESS power headroom, SOC energy margin, and PCC export/import headroom are still often not considered together. This paper therefore introduces a reserve deliverability formulation that combines power margin constraints, dual-endpoint SOC energy margin constraints, and PCC headroom constraints.

2.3. Degradation-Aware Modeling and Scenario Construction

Battery degradation is a major source of uncertainty in the life-cycle economics of BESS planning. The aging process is affected by the SOC trajectory, the depth of discharge, equivalent full cycles, calendar aging, temperature, and service-specific operating stress. If these effects are ignored, the economic value of storage may be overestimated, especially for applications involving frequent cycling. Lee and Kim proposed a degradation cost model for BESS scheduling that accounts for equivalent cycles and depth-of-discharge effects [7]. Park et al. reviewed degradation models for battery-based power service applications and pointed out that the choice of degradation model should be consistent with the operating characteristics of the target service [8]. Other recent studies have also reported that degradation representation can influence the design and scheduling of hybrid energy systems and BESS-based resources [29–33].

Recent studies have also extended energy storage operation toward multi-timescale and uncertainty-aware coordination. Mao et al. [34] developed a day-ahead, intra-day, and real-time scheduling framework that coordinates mobile energy storage, hydrogen storage, and air-conditioning cluster virtual storage according to their different temporal response characteristics. Mao and Jiao [35] modeled delay-tolerant batch computing loads as virtual batteries and proposed a two-stage Wasserstein distributionally robust co-scheduling framework in which flexible computing loads, gas turbines, physical energy storage, and renewable curtailment respond to wind and PV forecast errors. These studies demonstrate the value of coordinating physical and virtual flexibility across multiple operating stages under renewable uncertainty.

Including battery aging directly in dispatch optimization is possible, but detailed degradation models often introduce additional nonlinearities or temporal coupling. This difficulty becomes more pronounced when the dispatch problem is formulated as a multi-scenario MILP. One practical alternative is to optimize the dispatch first and then evaluate degradation as a post-processing correction to the planning result [9]. Following this idea, this study adopts a “dispatch optimization–rainflow degradation post-processing–capacity evaluation” procedure. The lower-level dispatch model remains linear and tractable, while the rainflow-based cycle aging cost is still included in the upper-level comparison of candidate BESS configurations.

The accuracy of degradation evaluation also depends on how renewable generation scenarios are represented. Storage operation is driven by chronological variations in renewable output, and SOC cycling may be underestimated when representative periods are overly smoothed. In particular, using clustering centroids directly as typical days can average out short-term ramps and local peak–valley variations, which may affect the optimized SOC trajectory and the subsequent rainflow-counting result. Studies on high-temporal-resolution planning and representative period selection have shown that storage expansion and operation are sensitive to temporal representation [10,15–17]. For this reason, degradation-aware BESS sizing requires representative scenarios that retain realistic intra-day fluctuation patterns without making the optimization problem unnecessarily large.

The above literature provides a useful foundation for BESS sizing, renewable integration, ancillary service participation, degradation evaluation, and scenario reduction.

However, several issues remain insufficiently addressed for renewable plant-level BESS sizing under prescribed reserve requirements: Ancillary services are often modeled as optional revenue streams, whereas reserve requirements used for planning evaluation need to be represented explicitly. Available reserve is also frequently estimated from rated power or instantaneous power headroom, without jointly considering SOC energy margins and PCC export/import limits. In addition, degradation evaluation is commonly separated from reserve-constrained sizing, and representative scenarios constructed from clustering centroids may suppress short-term renewable fluctuations that are relevant to SOC cycling. To address these issues, this study develops a reserve deliverability-constrained BESS sizing framework that integrates prescribed upward/downward reserve requirements, deliverable reserve constraints, real day-mapped typical scenarios, extreme stress test scenarios, and rainflow-based degradation post-processing.

3. Methodology

3.1. Framework Overview

This study develops a reserve deliverability-constrained BESS sizing framework for renewable power plants with cycling degradation considered in the capacity evaluation. The studied plant is treated not only as an energy supplier, but also as a grid-connected resource that should maintain prescribed upward and downward reserve capability under operational constraints. Accordingly, the sizing problem is formulated by linking renewable output variability, intra-day BESS operation, reserve requirements, PCC export/import limits, and degradation-related cost. The proposed procedure contains four main steps: scenario construction, operational dispatch, degradation assessment, and capacity evaluation.

Historical 15 min renewable generation data are first transformed into representative operating scenarios. Typical scenarios are used to evaluate expected economic performance, whereas extreme scenarios are retained for stress testing under boundary operating conditions. For each candidate BESS configuration, the intra-day operation of the renewable power plant, BESS, and external grid is optimized through a lower-level mixed-integer linear programming (MILP) model. The dispatch model accounts for energy trading, BESS charging and discharging, renewable curtailment, upward/downward reserve provision, and reserve shortage penalties. After dispatch optimization, rainflow counting is applied to the obtained SOC trajectories, and the resulting degradation cost is included only in the upper-level candidate comparison, rather than in the lower-level MILP objective. The overall architecture of the proposed framework is shown in Figure 1.

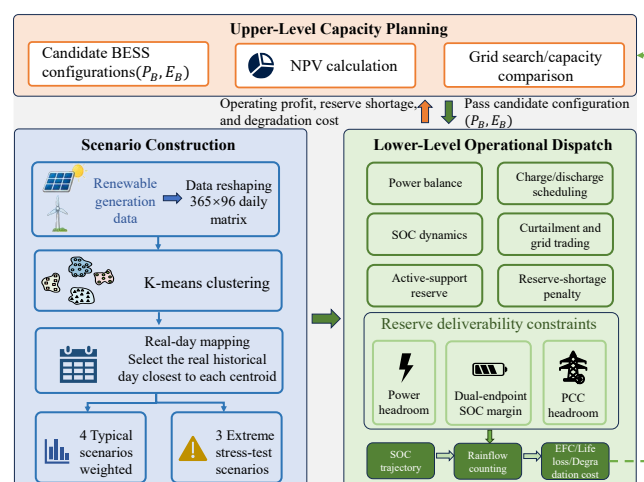


Figure 1. Overall framework of the proposed reserve deliverability-constrained BESS sizing method for renewable power plants.

3.2. Problem Statement and Basic Assumptions

The studied system is a renewable power plant equipped with a BESS and connected to the external grid through a point of common coupling (PCC). The renewable output may come from wind generation, PV generation, or their aggregation and is denoted by $P_{s,t}^{\text{RES}}$, where s represents the scenario index and t denotes the intra-day scheduling period. Through charging and discharging, the BESS adjusts the net power exchange between the plant and the grid while retaining upward and downward reserve capability within the prescribed SOC operating range.

For planning purposes, the renewable generation units and the BESS are represented as an aggregated single-node model at the PCC. This plant-level representation is used to evaluate BESS power and energy capacity, energy trading, renewable curtailment, and deliverable reserve capability at the plant–grid interface. The analysis focuses on the interaction among renewable output fluctuations, BESS capacity, upward/downward reserve provision, PCC headroom, and cycling degradation. Therefore, internal collection network power flow, internal losses, and multi-node voltage constraints are outside the scope of the present model. The main assumptions are listed below:

1. The renewable generation units and the BESS exchange power with the external grid through a common PCC. The internal collection network is not considered congested; therefore, the plant is represented by an aggregated single-node model.
2. The rated BESS power P_B and rated energy capacity E_B are determined in the upper-level capacity evaluation. For each candidate configuration, these two variables are fixed before solving the lower-level intra-day dispatch model.
3. Intra-day operation is described at a 15 min resolution. Each daily scheduling horizon consists of $T = 96$ periods, and the time step is $\Delta t = 0.25$ h.
4. BESS charging and discharging cannot occur simultaneously. Grid purchase and grid sale at the PCC are also mutually exclusive. The BESS energy level is required to return to its initial value at the end of each daily scheduling horizon.
5. The upward and downward reserve requirements are defined as proportional functions of real-time renewable output. This setting is used as a planning-level proxy for different grid-support responsibility levels associated with renewable generation, rather than as a representation of a specific ancillary service market product.
6. Battery degradation in the proposed formulation is assessed by applying rainflow cycle counting to the optimized SOC trajectory. The resulting degradation cost is included in the life-cycle economic evaluation of each candidate configuration. To avoid introducing nonlinear temporal coupling into the lower-level MILP, the proposed formulation adopts a decoupled “dispatch optimization–degradation post-processing–capacity evaluation” procedure. A separate throughput-based degradation-aware dispatch formulation is introduced only as a benchmark to evaluate the effect of allowing cycling considerations to influence the operational decisions.

The main symbols used in this paper are defined in Table 1.

3.3. Construction of Typical Scenarios and Extreme Scenarios

Renewable generation exhibits pronounced intra-day variability and seasonal changes. Using the full-year 15 min time series directly in the sizing model would require solving a large number of dispatch problems, which increases the computational burden. Representative-day methods based on clustering are often used to reduce this burden. However, when clustering centroids are directly adopted as typical days, short-term ramps and local peak–valley variations may be smoothed. This smoothing can affect the optimized SOC trajectories, reserve shortage estimation, and rainflow-based degradation assessment. To retain representative operating conditions while preserving boundary-case information,

this study constructs typical scenarios through clustering and separately extracts extreme scenarios from the historical data. The typical scenarios are used for expected-value economic evaluation, whereas the extreme scenarios are reserved for stress testing under boundary operating conditions.

Table 1. Definitions of main symbols.

Symbol	Description
$s \in \mathcal{S}$	Index and set of scenarios
$t \in \mathcal{T}$	Index of intra-day dispatch periods, $\mathcal{T} = \{1, \dots, T\}$
Δt	Dispatch time step
π_s	Probability weight of scenario s
$p_{s,t}^{\text{RES}}$	Renewable energy output in period t in scenario s
P_B	Rated power of the BESS
E_B	Rated energy capacity of the BESS
$p_{s,t}^{\text{ch}}, p_{s,t}^{\text{dis}}$	Charging and discharging power of the BESS
$p_{s,t}^{\text{sell}}, p_{s,t}^{\text{buy}}$	Power sold to and purchased from the grid by the plant
$p_{s,t}^{\text{curt}}$	Curtailed renewable power
$R_{s,t}^{\text{up}}, R_{s,t}^{\text{down}}$	Upward and downward reserve capacity
$D_{s,t}^{\text{up}}, D_{s,t}^{\text{down}}$	Upward and downward reserve requirements
$\Xi_{s,t}^{\text{up}}, \Xi_{s,t}^{\text{down}}$	Upward and downward reserve shortage
$E_{s,t}$	BESS energy level at the beginning of period t
$u_{s,t}^{\text{ch}}, u_{s,t}^{\text{dis}}$	Charging and discharging status variables of the BESS
$u_{s,t}^{\text{sell}}, u_{s,t}^{\text{buy}}$	Selling and purchasing status variables of the plant
$\eta^{\text{ch}}, \eta^{\text{dis}}$	Charging and discharging efficiencies of the BESS
$\text{SOC}_{\min}, \text{SOC}_{\max}$	Minimum and maximum SOC limits
SOC_0	Initial SOC of the BESS
λ_t^e	Energy price in period t
$\lambda^{\text{up}}, \lambda^{\text{down}}$	Compensation prices for upward and downward reserve
ρ^{curt}	Penalty coefficient for renewable curtailment
ρ^{short}	Penalty coefficient for reserve shortage
$p_{\max}^{\text{PCC,sell}}$	Maximum exported power at the PCC
$p_{\max}^{\text{PCC,buy}}$	Maximum imported power at the PCC
$\alpha^{\text{up}}, \alpha^{\text{down}}$	Ratios of upward and downward reserve requirements
τ	Required reserve deployment duration
DoD	Depth of discharge of a rainflow cycle
EFC	Equivalent full cycles
C_s^{deg}	Battery degradation cost in scenario s
C^{inv}	BESS investment cost
C^{OM}	Annual operation and maintenance cost
c_P, c_E	Power-related and energy-related investment cost coefficients
ξ^{OM}	Annual O&M cost rate
r	Discount rate
Y	Project lifetime
C_s^{thr}	Throughput-based dispatch penalty in scenario s (CNY)
c^{thr}	Unit battery throughput penalty (CNY/MWh-throughput)
$p_{s,t}^{\text{ch,k}}, p_{s,t}^{\text{dis,k}}$	Reserve-activated charging and discharging powers
$u_{s,t}^{\text{ch,k}}, u_{s,t}^{\text{dis,k}}$	Binary charging and discharging status variables under reserve activation
$\Delta E_{s,t}^k$	Incremental BESS energy change under reserve activation for $k \in \{\text{up}, \text{down}\}$
$\mathcal{E}_s^{\text{short}}$	Time-integrated upward and downward reserve shortage in scenario s
$\bar{\mathcal{E}}^{\text{short}}$	Probability-weighted daily reserve shortage over typical scenarios

Let the annual renewable generation sequence be

$$p^{\text{RES}} = [p_1^{\text{RES}}, p_2^{\text{RES}}, \dots, p_{35040}^{\text{RES}}], \quad (1)$$

where the temporal resolution is 15 min and the annual series contains $365 \times 96 = 35,040$ time points. The sequence is reshaped into the following daily sample matrix:

$$X = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,96} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,96} \\ \vdots & \vdots & \ddots & \vdots \\ x_{365,1} & x_{365,2} & \cdots & x_{365,96} \end{bmatrix}, \quad (2)$$

where the d -th row $x_d = [x_{d,1}, \dots, x_{d,96}]$ represents the 96-point renewable generation profile of day d .

K-means clustering is performed on the 365 daily profiles. Rather than treating the cluster centroids as synthetic representative days, the historical day with the shortest Euclidean distance to each centroid is selected as the corresponding typical scenario. In this way, each typical scenario remains an actual operating day from the original dataset. The mapped profiles therefore retain the original intra-day shape, including local ramps and peak–valley variations, and reduce the smoothing effect introduced by centroid averaging. This treatment provides more realistic chronological inputs for SOC trajectory optimization and rainflow-based degradation evaluation.

Given the number of clusters K , the K-means objective function is

$$\min_{\{\mathcal{C}_k\}_{k=1}^K, \{\mu_k\}_{k=1}^K} \sum_{k=1}^K \sum_{x_d \in \mathcal{C}_k} \|x_d - \mu_k\|_2^2, \quad (3)$$

where $\mathcal{C}_k$ denotes the k -th cluster and μ_k denotes the corresponding centroid. For each cluster, the mapped historical day is selected as follows:

$$d_k^* = \arg \min_{x_d \in \mathcal{C}_k} \|x_d - \mu_k\|_2. \quad (4)$$

The corresponding typical scenario is

$$P_k^{\text{typ}} = x_{d_k^*}, \quad k = 1, \dots, K. \quad (5)$$

The probability weight of each typical scenario is determined by the number of days in the corresponding cluster,

$$\pi_k = \frac{|\mathcal{C}_k|}{365}, \quad k = 1, \dots, K, \quad (6)$$

and satisfies

$$\sum_{k=1}^K \pi_k = 1. \quad (7)$$

Although typical scenarios represent frequently observed operating conditions, they may omit rare but operationally important days with sharp fluctuations or extremely high/low renewable output. To capture these boundary conditions, three extreme stress test scenarios are further extracted from the historical daily profiles.

The first stress test scenario is selected according to intra-day volatility. For day d , the volatility intensity is measured by the accumulated absolute variation between adjacent time periods:

$$V_d = \sum_{t=2}^{96} |x_{d,t} - x_{d,t-1}|. \quad (8)$$

The day with the largest volatility intensity is selected as the maximum-fluctuation scenario:

$$d_{\text{fluct}}^* = \arg \max_d V_d. \quad (9)$$

The other two stress test scenarios are selected according to the total daily renewable generation. The daily generation of day d is calculated as follows:

$$G_d = \sum_{t=1}^{96} x_{d,t} \Delta t. \quad (10)$$

The minimum-generation scenario is selected by

$$d_{\text{min}}^* = \arg \min_d G_d. \quad (11)$$

To avoid selecting invalid days caused by sensor failures or data anomalies, days with an excessive proportion of zero-output points or an abnormally low maximum output are excluded before determining the final minimum-generation scenario.

The maximum-generation scenario is selected by

$$d_{\text{max}}^* = \arg \max_d G_d. \quad (12)$$

The three extreme scenarios are assigned zero probability and are therefore excluded from the expected annual economic evaluation. They are used only to test the operational performance of candidate BESS configurations under boundary conditions. Algorithm 1 summarizes the overall scenario generation procedure.

Algorithm 1 Construction of typical scenarios and extraction of extreme scenarios

Require: Annual 15-min renewable generation sequence $\mathbf{p}^{\text{RES}}$, number of clusters K

Ensure: Typical scenario set $\mathcal{S}^{\text{typ}}$, extreme scenario set $\mathcal{S}^{\text{ext}}$

- 1: Reshape the annual output sequence into a 365×96 daily sample matrix $\mathbf{X}$
 - 2: Perform K-means clustering on $\mathbf{X}$ to obtain clusters $\mathcal{C}_k$ and centroids μ_k
 - 3: **for** $k = 1$ to K **do**
 - 4: Select the real historical day closest to the centroid according to Equation (4)
 - 5: Calculate the scenario weight $\pi_k = |\mathcal{C}_k|/365$
 - 6: **end for**
 - 7: Extract the maximum-fluctuation day according to Equation (8)
 - 8: Extract the minimum- and maximum-generation days according to Equation (10)
 - 9: Assign cluster probabilities to typical scenarios and zero probability to extreme scenarios
 - 10: **return** $\mathcal{S}^{\text{typ}}$, $\mathcal{S}^{\text{ext}}$
-

3.4. Coupled Sizing–Operation Modeling Framework

The proposed method links capacity selection with scenario-based operational dispatch. At the upper level, candidate BESS configurations are compared in terms of life-cycle economic performance, and the rated power P_B and energy capacity E_B are selected from predefined candidate sets. At the lower level, each candidate configuration is fixed and evaluated through intra-day dispatch in the representative scenarios. The dispatch results provide the operating profit, renewable curtailment, reserve shortage, and SOC trajectory required for the upper-level evaluation.

Battery degradation is evaluated after dispatch rather than embedded directly in the MILP model. This treatment is adopted because rainflow counting depends on the turning points and cycle structure of the SOC trajectory, which would introduce nonlinear temporal coupling if modeled within the dispatch optimization. Therefore, the lower-level

problem is kept as a tractable MILP, while the optimized SOC trajectory is used for rainflow-based degradation post-processing. The resulting degradation cost is then included in the life-cycle comparison of candidate BESS configurations.

For a given capacity configuration (P_B, E_B) , the operating profit in scenario s is calculated from the energy revenue, reserve revenue, curtailment penalty, and reserve shortage penalty:

$$R_s^{\text{op}}(P_B, E_B) = R_s^{\text{e}} + R_s^{\text{res}} - C_s^{\text{curt}} - C_s^{\text{short}}, \quad (13)$$

where R_s^{e} denotes the energy revenue, R_s^{res} denotes the upward/downward reserve revenue, C_s^{curt} denotes the curtailment penalty, and C_s^{short} denotes the reserve shortage penalty.

Only typical scenarios are used in the expected annual economic evaluation. The scenario-weighted daily operating profit and degradation cost are given by

$$\bar{R}^{\text{op}}(P_B, E_B) = \sum_{s \in \mathcal{S}^{\text{typ}}} \pi_s R_s^{\text{op}}(P_B, E_B), \quad (14)$$

$$\bar{C}^{\text{deg}}(P_B, E_B) = \sum_{s \in \mathcal{S}^{\text{typ}}} \pi_s C_s^{\text{deg}}(P_B, E_B). \quad (15)$$

The annual net profit is then obtained by annualizing the expected daily operating profit and degradation cost and subtracting the annual operation and maintenance cost:

$$R_{\text{net}}^{\text{annual}} = 365 [\bar{R}^{\text{op}}(P_B, E_B) - \bar{C}^{\text{deg}}(P_B, E_B)] - C^{\text{OM}}. \quad (16)$$

The investment cost consists of power-related and energy-related components,

$$C^{\text{inv}} = c_P P_B + c_E E_B, \quad (17)$$

and the annual operation and maintenance cost is calculated as a fixed proportion of the investment cost:

$$C^{\text{OM}} = \zeta^{\text{OM}} C^{\text{inv}}, \quad (18)$$

where ζ^{OM} is the annual O&M cost rate.

The life-cycle economic performance of each candidate configuration is measured by the net present value:

$$\text{NPV}(P_B, E_B) = -C^{\text{inv}} + \sum_{y=1}^Y \frac{R_{\text{net}}^{\text{annual}}}{(1+r)^y}, \quad (19)$$

where Y is the project lifetime and r is the discount rate.

The upper-level sizing problem is therefore written as follows:

$$\max_{P_B, E_B} \text{NPV}(P_B, E_B), \quad (20)$$

$$\text{s.t. } P_B \in \mathcal{P}, \quad E_B = P_B h, \quad h \in \mathcal{H}, \quad (21)$$

where $\mathcal{P}$ denotes the candidate set of rated BESS power, and $\mathcal{H}$ denotes the candidate set of storage duration. The problem is solved by discrete grid search over (P_B, h) . This is consistent with practical capacity selection from finite power-rating and duration options, and it keeps the upper-level comparison transparent and reproducible.

3.5. Lower-Level Operational Dispatch Model for the Renewable Power Plant, BESS, and Grid

For a given BESS configuration (P_B, E_B) and scenario s , the lower-level model optimizes the intra-day operation of the renewable power plant, the BESS, and the external

grid. The problem is formulated as an MILP. Its objective is to maximize the single-day operating profit while satisfying the plant-level power balance, BESS operating limits, PCC exchange constraints, and reserve deliverability requirements.

The operating profit in scenario s consists of the energy trading revenue, upward/-downward reserve revenue, curtailment penalty, and reserve shortage penalty:

$$\max R_s^{\text{op}} = \sum_{t \in \mathcal{T}} \left[\lambda_t^e (P_{s,t}^{\text{sell}} - P_{s,t}^{\text{buy}}) \Delta t + \lambda^{\text{up}} R_{s,t}^{\text{up}} \Delta t + \lambda^{\text{down}} R_{s,t}^{\text{down}} \Delta t - \rho^{\text{curt}} P_{s,t}^{\text{curt}} \Delta t - \rho^{\text{short}} (\Xi_{s,t}^{\text{up}} + \Xi_{s,t}^{\text{down}}) \Delta t \right]. \quad (22)$$

Here, λ_t^e is the time-varying energy price. The coefficients λ^{up} and λ^{down} are modeled as the compensation for maintaining upward and downward reserve capacity available during each dispatch interval. The model does not separately represent balancing energy activation or the associated activation energy settlement. The coefficient ρ^{curt} represents the planning-level opportunity cost of renewable curtailment, whereas ρ^{short} is an administrative reliability penalty assigned to insufficient prescribed reserve capability. It should not be interpreted as the imbalance settlement price or the shortage price of a specific balancing market. A relatively large value of ρ^{short} is adopted to discourage reserve shortages while preserving model feasibility under severe operating conditions. The parameter basis and sensitivity settings are further discussed in Sections 4.1 and 4.8.

The plant-level power balance requires renewable generation, BESS discharging power, and grid purchase to be allocated among grid sales, BESS charging, and renewable curtailment:

$$P_{s,t}^{\text{RES}} + P_{s,t}^{\text{dis}} + P_{s,t}^{\text{buy}} = P_{s,t}^{\text{sell}} + P_{s,t}^{\text{ch}} + P_{s,t}^{\text{curt}}, \quad \forall t \in \mathcal{T}. \quad (23)$$

The BESS charging and discharging powers are limited by the rated power P_B and their corresponding binary status variables:

$$0 \leq P_{s,t}^{\text{ch}} \leq u_{s,t}^{\text{ch}} P_B, \quad \forall t \in \mathcal{T}, \quad (24)$$

$$0 \leq P_{s,t}^{\text{dis}} \leq u_{s,t}^{\text{dis}} P_B, \quad \forall t \in \mathcal{T}, \quad (25)$$

$$u_{s,t}^{\text{ch}} + u_{s,t}^{\text{dis}} \leq 1, \quad u_{s,t}^{\text{ch}}, u_{s,t}^{\text{dis}} \in \{0, 1\}, \quad \forall t \in \mathcal{T}. \quad (26)$$

Equation (26) enforces the mutually exclusive charging and discharging states.

The BESS energy state is modeled in nodal form. Let $E_{s,t}$ denote the stored energy at the beginning of period t . The intertemporal energy transition is

$$E_{s,t+1} = E_{s,t} + \eta^{\text{ch}} P_{s,t}^{\text{ch}} \Delta t - \frac{P_{s,t}^{\text{dis}}}{\eta^{\text{dis}}} \Delta t, \quad \forall t \in \mathcal{T}. \quad (27)$$

The stored energy is constrained by the allowable SOC range:

$$\text{SOC}_{\min} E_B \leq E_{s,t} \leq \text{SOC}_{\max} E_B, \quad \forall t = 1, \dots, T+1. \quad (28)$$

The initial energy level is set as

$$E_{s,1} = \text{SOC}_0 E_B. \quad (29)$$

To prevent artificial energy transfer across representative daily horizons, the terminal energy level is required to return to the initial value:

$$E_{s,T+1} = E_{s,1}. \quad (30)$$

Controlled 48 h benchmarking is further conducted in Section 4.5 to evaluate the effect of removing the intermediate 24 h SOC reset while retaining energy neutrality over the complete 48 h horizon.

The renewable power plant exchanges power with the external grid through the PCC. The export and import powers are limited by the PCC capacity and their binary status variables:

$$0 \leq P_{s,t}^{\text{sell}} \leq u_{s,t}^{\text{sell}} P_{\text{max}}^{\text{PCC,sell}}, \quad \forall t \in \mathcal{T}, \quad (31)$$

$$0 \leq P_{s,t}^{\text{buy}} \leq u_{s,t}^{\text{buy}} P_{\text{max}}^{\text{PCC,buy}}, \quad \forall t \in \mathcal{T}, \quad (32)$$

$$u_{s,t}^{\text{sell}} + u_{s,t}^{\text{buy}} \leq 1, \quad u_{s,t}^{\text{sell}}, u_{s,t}^{\text{buy}} \in \{0, 1\}, \quad \forall t \in \mathcal{T}. \quad (33)$$

Equation (33) avoids simultaneous grid purchase and sale at the PCC.

When external grid purchase is not allowed in a specific setting, the import variables can be fixed to zero:

$$P_{s,t}^{\text{buy}} = 0, \quad u_{s,t}^{\text{buy}} = 0, \quad \forall t \in \mathcal{T}. \quad (34)$$

3.6. Reserve Requirements and Deliverability Constraints

In addition to energy delivery, the renewable power plant is evaluated with respect to its ability to maintain upward and downward reserve capability. The required reserve is specified as a proportion of the renewable output:

$$D_{s,t}^{\text{up}} = \alpha^{\text{up}} P_{s,t}^{\text{RES}}, \quad D_{s,t}^{\text{down}} = \alpha^{\text{down}} P_{s,t}^{\text{RES}}, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (35)$$

Here, $D_{s,t}^{\text{up}}$ and $D_{s,t}^{\text{down}}$ denote the upward and downward reserve requirements, respectively. The proportional formulation is adopted as a transparent planning-level parameterization for representing the different levels of prescribed active support responsibility associated with renewable generation, rather than as a direct representation of a specific FCR, aFRR, or mFRR product. In European power system operation, FRR capacity is generally dimensioned at the load frequency control block level using historical system imbalances, probabilistic reliability criteria, and dimensioning incidents [36]. Studies conducted for the Belgian system have further shown that operating reserve requirements can be dynamically determined according to the estimated risk of system imbalance [37], while probabilistic forecast-based methods explicitly incorporate forecast uncertainty into reserve dimensioning [38]. Accordingly, the ratios α^{up} and α^{down} are treated in this study as scenario parameters for examining how different prescribed reserve responsibility levels affect BESS capacity selection.

The scheduled upward and downward reserve capacities should cover the corresponding requirements. Non-negative reserve shortage variables are introduced to preserve feasibility under severe operating conditions:

$$R_{s,t}^{\text{up}} + \Xi_{s,t}^{\text{up}} \geq D_{s,t}^{\text{up}}, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (36)$$

$$R_{s,t}^{\text{down}} + \Xi_{s,t}^{\text{down}} \geq D_{s,t}^{\text{down}}, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (37)$$

The non-negative variables $\Xi_{s,t}^{\text{up}}$ and $\Xi_{s,t}^{\text{down}}$ measure insufficient upward and downward reserve, respectively, and their values are penalized in the lower-level dispatch objective.

For scenario s , the time-integrated bidirectional reserve shortage is calculated as

$$\mathcal{E}_s^{\text{short}} = \sum_{t \in \mathcal{T}} (\Xi_{s,t}^{\text{up}} + \Xi_{s,t}^{\text{down}}) \Delta t. \quad (38)$$

For the typical-scenario evaluation, the probability-weighted daily reserve shortage is

$$\bar{\mathcal{E}}^{\text{short}} = \sum_{s \in \mathcal{S}^{\text{typ}}} \pi_s \mathcal{E}_s^{\text{short}}. \quad (39)$$

The metric is reported in MWh-eq/day and represents the time-integrated sum of upward and downward reserve shortages. For an individual stress test scenario, $\mathcal{E}_s^{\text{short}}$ is reported directly without probability weighting. To examine the effect of strictly enforcing the prescribed reserve requirements, a hard-reserve benchmark is additionally formulated by fixing the upward and downward reserve shortage variables to zero:

$$\Xi_{s,t}^{\text{up}} = \Xi_{s,t}^{\text{down}} = 0, \quad \forall s \in \mathcal{S}^{\text{typ}}, t \in \mathcal{T}. \quad (40)$$

Under this formulation, a candidate configuration is classified as hard-infeasible if its dispatch problem is infeasible in any typical scenario. The soft-shortage formulation is retained as the main formulation because it preserves feasibility and quantifies the magnitude of reserve insufficiency under capacity-limited or stress conditions. The hard-reserve benchmark is used to evaluate whether the capacity selection result remains valid when zero reserve shortage is strictly required in all typical scenarios and dispatch periods.

Reserve provision is first constrained by the BESS power margin. In the main sizing formulation, upward reserve is represented through additional discharging capability, whereas downward reserve is represented through additional charging capability. The scheduled power and reserved power must therefore satisfy the following:

$$P_{s,t}^{\text{dis}} + R_{s,t}^{\text{up}} \leq P_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (41)$$

$$P_{s,t}^{\text{ch}} + R_{s,t}^{\text{down}} \leq P_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (42)$$

Equations (41) and (42) retain the linear structure of the baseline dispatch model; however, they do not explicitly credit the upward modulation obtained by reducing scheduled charging or the downward modulation obtained by reducing scheduled discharging. Consequently, the main formulation provides a conservative representation of the complete net power modulation capability of the BESS.

The power margin alone is insufficient for evaluating deliverable reserve; the reserved power must also be sustainable for the required deployment duration τ without violating the SOC limits. Here, τ denotes the assumed sustainment horizon over which the scheduled reserve must remain deliverable with respect to the BESS energy limits. It is treated as a planning parameter and is not intended to reproduce the activation duration requirement of a specific FCR, aFRR, or mFRR product. Because the BESS energy state changes during each scheduling interval, the reserve energy margin is checked at both the beginning and the end of the interval.

For upward reserve, sufficient dischargeable energy must be retained:

$$E_{s,t} - \frac{R_{s,t}^{\text{up}} \tau}{\eta^{\text{dis}}} \geq \text{SOC}_{\min} E_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (43)$$

$$E_{s,t+1} - \frac{R_{s,t}^{\text{up}} \tau}{\eta^{\text{dis}}} \geq \text{SOC}_{\min} E_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (44)$$

For downward reserve, sufficient charging space must remain available:

$$E_{s,t} + R_{s,t}^{\text{down}} \eta^{\text{ch}} \tau \leq \text{SOC}_{\max} E_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (45)$$

$$E_{s,t+1} + R_{s,t}^{\text{down}} \eta^{\text{ch}} \tau \leq \text{SOC}_{\max} E_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (46)$$

Equations (43)–(46) impose dual-endpoint SOC margin requirements. They reduce the possibility of a scheduling reserve that appears feasible before the scheduled charging or discharging action but becomes infeasible after the corresponding energy transition.

Reserve deliverability is further restricted by the PCC. Even when the BESS has adequate power and energy margins, the scheduled reserve may not be deliverable if the baseline power exchange already occupies the available PCC capacity. Therefore, the net plant–grid exchange following reserve activation is constrained by the PCC export and import limits.

For upward reserve, the net export after reserve activation is limited by

$$P_{s,t}^{\text{sell}} - P_{s,t}^{\text{buy}} + R_{s,t}^{\text{up}} \leq P_{\max}^{\text{PCC,sell}}, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (47)$$

whereas, for downward reserve, the net import after reserve activation is limited by

$$P_{s,t}^{\text{buy}} - P_{s,t}^{\text{sell}} + R_{s,t}^{\text{down}} \leq P_{\max}^{\text{PCC,buy}}, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (48)$$

Together, Equations (41)–(48) define the reserve deliverability constraints used in the main sizing formulation. The constraint set jointly checks the BESS power margin, dual-endpoint SOC energy margin, and PCC export/import headroom, thereby avoiding an evaluation of reserve capability based only on rated power or instantaneous BESS power headroom.

To assess the conservativeness of the baseline reserve power representation, an alternative flexible net power modulation formulation is additionally constructed for sensitivity analysis. This formulation explicitly represents the charging and discharging states of the BESS following upward or downward reserve activation. Let $P_{s,t}^{\text{ch,up}}$ and $P_{s,t}^{\text{dis,up}}$ denote the reserve-activated charging and discharging powers under upward reserve deployment, respectively, and let $P_{s,t}^{\text{ch,down}}$ and $P_{s,t}^{\text{dis,down}}$ denote the corresponding powers under downward reserve deployment.

The reserve-activated net BESS power must differ from the baseline net power by the scheduled reserve capacity. For upward reserve activation, the following relationship is imposed:

$$P_{s,t}^{\text{dis,up}} - P_{s,t}^{\text{ch,up}} = P_{s,t}^{\text{dis}} - P_{s,t}^{\text{ch}} + R_{s,t}^{\text{up}}, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (49)$$

For downward reserve activation, the corresponding relationship is

$$P_{s,t}^{\text{dis,down}} - P_{s,t}^{\text{ch,down}} = P_{s,t}^{\text{dis}} - P_{s,t}^{\text{ch}} - R_{s,t}^{\text{down}}, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (50)$$

The reserve-activated charging and discharging powers are constrained by the rated BESS power:

$$0 \leq P_{s,t}^{\text{ch,k}} \leq u_{s,t}^{\text{ch,k}} P_B, \quad (51)$$

$$0 \leq P_{s,t}^{\text{dis,k}} \leq u_{s,t}^{\text{dis,k}} P_B, \quad (52)$$

for all $s \in \mathcal{S}$, $t \in \mathcal{T}$, and $k \in \{\text{up}, \text{down}\}$. The reserve-activated charging and discharging states remain mutually exclusive:

$$u_{s,t}^{\text{ch,k}} + u_{s,t}^{\text{dis,k}} \leq 1, \quad u_{s,t}^{\text{ch,k}}, u_{s,t}^{\text{dis,k}} \in \{0, 1\}, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, k \in \{\text{up}, \text{down}\}. \quad (53)$$

Compared with the baseline schedule, the incremental BESS energy change caused by upward reserve activation over the required deployment duration τ is defined as

$$\Delta E_{s,t}^{\text{up}} = \left[\eta^{\text{ch}} (P_{s,t}^{\text{ch,up}} - P_{s,t}^{\text{ch}}) - \frac{P_{s,t}^{\text{dis,up}} - P_{s,t}^{\text{dis}}}{\eta^{\text{dis}}} \right] \tau, \quad (54)$$

and the corresponding incremental energy change under downward reserve activation is

$$\Delta E_{s,t}^{\text{down}} = \left[\eta^{\text{ch}} (P_{s,t}^{\text{ch,down}} - P_{s,t}^{\text{ch}}) - \frac{P_{s,t}^{\text{dis,down}} - P_{s,t}^{\text{dis}}}{\eta^{\text{dis}}} \right] \tau. \quad (55)$$

The reserve-activated energy state is checked at both endpoints of each scheduling interval. For upward reserve activation, sufficient stored energy must remain above the minimum SOC limit:

$$E_{s,t} + \Delta E_{s,t}^{\text{up}} \geq \text{SOC}_{\min} E_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (56)$$

$$E_{s,t+1} + \Delta E_{s,t}^{\text{up}} \geq \text{SOC}_{\min} E_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (57)$$

For downward reserve activation, sufficient charging space must remain below the maximum SOC limit:

$$E_{s,t} + \Delta E_{s,t}^{\text{down}} \leq \text{SOC}_{\max} E_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}, \quad (58)$$

$$E_{s,t+1} + \Delta E_{s,t}^{\text{down}} \leq \text{SOC}_{\max} E_B, \quad \forall s \in \mathcal{S}, \forall t \in \mathcal{T}. \quad (59)$$

The PCC headroom constraints in Equations (47) and (48) remain applicable because upward reserve activation increases the plant net export by $R_{s,t}^{\text{up}}$, whereas downward reserve activation decreases the net export by $R_{s,t}^{\text{down}}$. The flexible formulation therefore permits upward reserve to be delivered by reducing charging, increasing discharging, or transitioning from charging to discharging. Similarly, downward reserve can be delivered by reducing discharging, increasing charging, or transitioning from discharging to charging.

The baseline formulation is retained as the main sizing formulation because it has a simpler model structure and lower computational burden. The flexible net power modulation formulation is used only in Section 4.5 to quantify whether the conservative reserve power representation materially affects the selected BESS configuration.

3.7. Battery Degradation Assessment Based on Rainflow Cycle Counting

Battery cycling degradation in the proposed formulation is evaluated from the optimized SOC trajectory obtained after the lower-level dispatch. Compared with a simple energy-throughput approximation, rainflow counting can reflect the depth and structure of SOC cycles, which are important for estimating cycle-related aging. Therefore, the proposed formulation applies rainflow cycle counting as a post-processing step in the candidate-level capacity evaluation.

For scenario s , the optimized SOC trajectory is written as

$$\text{SOC}_s = [\text{SOC}_{s,1}, \text{SOC}_{s,2}, \dots, \text{SOC}_{s,T+1}], \quad (60)$$

and is decomposed by rainflow counting into a set of cycles:

$$\mathcal{I}_s = \{ (D_0 D_{s,i}, n_{s,i}, \overline{\text{SOC}_{s,i}}) \}_{i=1}^{N_s}, \quad (61)$$

where $DoD_{s,i}$ denotes the depth of the i -th cycle, $n_{s,i}$ denotes the corresponding cycle count, and $\overline{SOC}_{s,i}$ denotes the average SOC of the cycle. A full cycle corresponds to $n_{s,i} = 1$, whereas a half cycle corresponds to $n_{s,i} = 0.5$. In this study, the degradation model focuses on cycle depth effects and does not explicitly represent C-rate, temperature, calendar aging, or average-SOC effects.

The cycle life at a given depth of discharge is described by the following empirical function:

$$N^{cyc}(DoD) = N_{ref} \left(\frac{DoD_{ref}}{\max(DoD, DoD_{min})} \right)^{\beta}, \quad (62)$$

where N_{ref} is the cycle life at the reference depth DoD_{ref} , DoD_{min} is introduced to avoid unrealistically large cycle life values at very small cycle depths, and β is the degradation sensitivity coefficient.

The life loss associated with cycle i in scenario s is calculated as follows:

$$L_{s,i} = \frac{n_{s,i}}{N^{cyc}(DoD_{s,i})}. \quad (63)$$

The scenario-level life loss is then obtained by summing the losses of all identified cycles:

$$L_s = \sum_{i=1}^{N_s} L_{s,i}. \quad (64)$$

The equivalent full cycles in scenario s are calculated as follows:

$$EFC_s = \sum_{i=1}^{N_s} n_{s,i} DoD_{s,i}. \quad (65)$$

Finally, the degradation cost is estimated by multiplying the scenario-level life loss by the energy capacity-related investment cost:

$$C_s^{deg} = c_E E_B L_s. \quad (66)$$

Here, $c_E E_B$ is used as a proxy for the replacement value of the battery energy component.

The adopted degradation model is empirical and cycle depth-based. It is intended to capture the effect of different SOC cycling patterns generated by dispatch decisions, rather than to reproduce detailed electrochemical aging mechanisms. C-rate, temperature, calendar aging, and average-SOC effects are not explicitly modeled. As a result, configurations with the same energy capacity and similar rainflow cycle depth distributions may show similar life loss estimates even when their rated powers differ. This simplification keeps the degradation assessment transparent and reproducible while leaving more detailed aging models for future extensions.

To examine the effect of allowing cycling considerations to influence the operational decisions, an additional throughput-based degradation-aware dispatch formulation is introduced as a tractable benchmark. For scenario s , the battery throughput penalty is defined as follows:

$$C_s^{thr} = c^{thr} \sum_{t \in \mathcal{T}} (P_{s,t}^{ch} + P_{s,t}^{dis}) \Delta t, \quad (67)$$

where c^{thr} denotes the unit throughput penalty coefficient in CNY/MWh-throughput. For this degradation-aware dispatch benchmark, the lower-level objective is modified as follows:

$$\max(R_s^{op} - C_s^{thr}), \quad (68)$$

where R_s^{op} denotes the original scenario-level operating objective before the throughput penalty is introduced.

The throughput penalty coefficient is calibrated from the energy-related battery investment cost and the reference cycle life parameters as follows:

$$c^{\text{thr}} = \frac{c_E}{2N_{\text{ref}}DoD_{\text{ref}}}, \quad (69)$$

which gives $c^{\text{thr}} = 125$ CNY/MWh-throughput under the parameter settings adopted in this study. The factor of two accounts for the inclusion of both charging and discharging energy in the throughput metric.

The throughput penalty is used only as a linear dispatch surrogate to discourage excessive charging and discharging throughput; it is not interpreted as an independent physical degradation cost. After the degradation-aware dispatch problem is solved, the resulting SOC trajectory is evaluated using the same rainflow-based degradation model defined in Equations (60)–(66). For a consistent candidate-level comparison with the proposed formulation, the NPV of the degradation-aware dispatch benchmark is calculated using the rainflow degradation cost C_s^{deg} . Accordingly, the throughput penalty C_s^{thr} is not separately deducted again in the candidate-level NPV, thereby avoiding the simultaneous accounting of the throughput surrogate and the rainflow degradation estimate as independent physical costs.

3.8. Solution Method for the Upper-Level BESS Sizing Problem

The upper-level problem determines the rated BESS power P_B and energy capacity E_B . Since practical BESS projects are often selected from finite power ratings and storage duration options, the upper-level sizing problem is solved by enumerating a predefined set of candidate configurations. This discrete search is suitable for the present study because the focus is on comparing capacity options under the same reserve deliverability and rainflow-based degradation evaluation framework, rather than on developing a new continuous optimization algorithm.

Let $\mathcal{P}$ denote the candidate set of rated BESS power and $\mathcal{H}$ denote the candidate set of storage duration. For each $P_B \in \mathcal{P}$ and $h \in \mathcal{H}$, the corresponding energy capacity is defined as follows:

$$E_B = P_B h, \quad P_B \in \mathcal{P}, h \in \mathcal{H}. \quad (70)$$

Each candidate configuration (P_B, E_B) is evaluated through the same scenario-based procedure. The lower-level MILP dispatch model is first solved for all typical scenarios. The optimized operating profit and SOC trajectory are then used to calculate rainflow-based life loss, EFC, and degradation cost. After scenario weighting and annualization, the operating profit and degradation cost are combined with investment cost, O&M cost, discount rate, and project lifetime to obtain the NPV. The candidate with the largest NPV is reported as the best-performing configuration within the predefined candidate set, rather than as a global optimum over continuous BESS power and energy capacities. Algorithm 2 summarizes the solution procedure.

The discrete search converts the upper-level sizing problem into a finite comparison of candidate configurations. This treatment keeps the evaluation process reproducible and makes the effect of reserve requirements, PCC constraints, and degradation cost easier to trace across different capacity options. If finer capacity granularity is required, the same evaluation structure can be combined with continuous capacity optimization, heuristic search, or stochastic planning methods.

Algorithm 2 Solution procedure for reserve deliverability-constrained BESS sizing considering cycling degradation

Require: Typical scenario set $\mathcal{S}^{\text{typ}}$, candidate power set $\mathcal{P}$, candidate duration set $\mathcal{H}$

Ensure: Selected BESS configuration (P_B^*, E_B^*)

```

1: Initialize the result set  $\mathcal{R} \leftarrow \emptyset$ 
2: for each  $P_B \in \mathcal{P}$  do
3:   for each  $h \in \mathcal{H}$  do
4:     Set  $E_B = P_B h$ 
5:     for each  $s \in \mathcal{S}^{\text{typ}}$  do
6:       Solve the lower-level MILP dispatch model to obtain  $R_s^{\text{op}}$  and  $\text{SOC}_s$ 
7:       Perform rainfall counting on  $\text{SOC}_s$ 
8:       Calculate  $C_s^{\text{deg}}$  and  $\text{EFC}_s$ 
9:     end for
10:    Calculate the expected daily operating profit  $\bar{R}^{\text{op}}$  and expected degradation cost  $\bar{C}^{\text{deg}}$ 
11:    Calculate the NPV based on investment cost, O&M cost, discount rate, and project lifetime
12:    Store  $(P_B, E_B, h, \text{NPV})$  in  $\mathcal{R}$ 
13:   end for
14: end for
15: Select the configuration with the maximum NPV in  $\mathcal{R}$ 
16: return  $(P_B^*, E_B^*)$ 
  
```

4. Experimental Results and Analysis

4.1. Experimental Data and Scenario Construction

The numerical study uses the 2023 nationally aggregated Belgian wind and photovoltaic generation data at a 15 min resolution, yielding $365 \times 96 = 35,040$ observations. The records are aligned to a complete local-time index; wind observations are aggregated by timestamp and forward-filled; PV records are deduplicated and interpolated; and negative dispatch inputs are clipped to zero. Because the data represent a geographically aggregated renewable portfolio rather than a specific plant, the model is interpreted at an equivalent single PCC, and the 10,000 MW PCC limit and reported BESS capacities are defined on the same aggregated system scale. Although real-day mapping avoids the additional smoothing caused by directly using cluster centroids, it cannot recover site-level variability absent from the source data. The case study therefore evaluates the behavior of the proposed framework rather than prescribing the design of an individual renewable power plant.

Tables 2 and 3 summarize the experimental settings and model parameters. The MILP models are implemented in Python 3.9.18 and solved using Gurobi 12.0.3 with a MIPGap tolerance of 10^{-4} and a 60 s time limit per dispatch instance. K-means is applied to the unscaled 365×96 daily profiles using k-means++ initialization, $n_{\text{init}} = 50$, $\text{random_state} = 2026$, $\text{max_iter} = 1000$, and $\text{tol} = 10^{-4}$. Cycling degradation is evaluated from all $T + 1$ per-unit SOC nodes using `rainflow` 3.2.0, with half cycles retained and $\text{DoD}_{\text{min}} = 0.05$ used only as the cycle life curve floor. Computations are performed on an Intel Core i9-13900HX computer with 16 GB RAM. The standard 96-period B4 model contains 1345 variables, including 384 binaries, and 1924 linear constraints. Across the 24 h instances with finite recorded runtimes, including the four time-limited cases, the mean runtime is 0.114 s. Four nonleading 24 h instances reach the 60 s solver time limit and return final optimality gaps of up to 1.96×10^{-4} ; the remaining normally terminated feasible instances satisfy the prescribed MIPGap tolerance of 10^{-4} . Sixteen auxiliary 48 h instances also reach the time limit, with final optimality gaps of up to 3.30×10^{-4} . None of these time-limited instances affects the top-five candidate rankings, and all selected configurations are independently re-solved to optimality.

Table 2. Main experimental settings.

Item	Setting
Time resolution	15 min
Dispatch periods per day	$T = 96$
Energy-state nodes	$T + 1 = 97$
Typical scenarios	4, used for probability-weighted NPV evaluation
Stress test scenarios	3, zero probability and used only for stress testing
Baseline reserve ratio	$\alpha^{\text{up}} = \alpha^{\text{down}} = 0.05$
Ablation reserve ratio	$\alpha^{\text{up}} = \alpha^{\text{down}} = 0.15$
Reserve ratio sensitivity	$\alpha \in \{0.05, 0.15, 0.25, 0.35\}$
Baseline PCC export/import limit	10,000/10,000 MW
PCC export limit sensitivity	$\{10,000, 6000, 5000, 4000\}$ MW
Candidate BESS power set	$\{300, 500, 800, 1000, 1500\}$ MW
Candidate duration set	$\{1, 2, 4\}$ h
BESS energy capacity	$E_B = P_B \times h$
Solver	Gurobi
MIPGap	10^{-4}
Time limit	60 s per dispatch instance

Table 3. Economic, technical, and degradation parameters.

Parameter	Value	Unit
Power investment cost c_P	1.5×10^6	CNY/MW
Energy investment cost c_E	1.2×10^6	CNY/MWh
Annual O&M rate ζ^{OM}	0.02	–
Project lifetime Y	10	years
Discount rate r	0.08	–
Valley energy price	250	CNY/MWh
Flat energy price	350	CNY/MWh
Peak energy price	600	CNY/MWh
Upward reserve price λ^{up}	10	CNY/(MW·h)
Downward reserve price λ^{down}	10	CNY/(MW·h)
Curtailed penalty ρ^{curt}	1000	CNY/MWh
Reserve shortage penalty ρ^{short}	5000	CNY/(MW·h)
Charging efficiency η^{ch}	0.95	–
Discharging efficiency η^{dis}	0.95	–
Initial SOC SOC_0	0.50	p.u.
SOC range	$[0.10, 0.90]$	p.u.
Reserve deployment duration τ	0.25	h
Rainflow cycle life N_{ref}	6000	cycles
Reference DoD D_{ref}	0.8	p.u.
Minimum DoD D_{min}	0.05	p.u.
Degradation exponent β	1.5	–

The investment cost model separates the power-related coefficient c_P from the energy-related coefficient c_E , consistent with utility-scale BESS cost assessment methods that distinguish battery energy costs from inverter, balance-of-system, and other power-related costs [2]. The adopted values are case study assumptions for the equivalent aggregated system planning case rather than quotations for a specific commercial project. The Belgian wholesale market report [39] provides the market context, whereas the adopted valley, flat, and peak electricity prices constitute a stylized time-of-use profile rather than a reconstruction of the complete historical day-ahead price sequence. The reserve prices λ^{up} and λ^{down} represent availability compensation rather than activated balancing energy payments [40]; ρ^{curt} represents a planning-level opportunity cost of renewable curtailment [41]; and ρ^{short} is an administrative reliability penalty rather than a market imbalance settlement price [42]. Because these coefficients are case-dependent, their effects are examined in Section 4.8.

The annual series is reshaped into a 365×96 daily profile matrix, and the number of clusters is evaluated using SSE, the Silhouette Score, and the Davies–Bouldin Index, as shown in Figure 2. Although $K = 2$ provides favorable cluster separation indices, increasing K from 2 to 4 reduces the SSE by approximately 35.7% and retains greater operational diversity at a manageable computational cost. To examine sizing stability, the original 15 candidate configurations are evaluated for $K \in \{2, 3, 4, 5, 6\}$ under $\alpha = 0.05$ and 0.15. As shown in Table 4, all tested values of K select 300 MW/300 MWh/1 h at $\alpha = 0.05$ and 1000 MW/1000 MWh/1 h at $\alpha = 0.15$. However, $K = 2$ underestimates the annual profile standard deviation, accumulated ramping, and maximum ramp by 7.87%, 10.26%, and 26.55%. Therefore, $K = 4$ is retained as a case-specific compromise among scenario diversity, sizing stability, and computational burden.

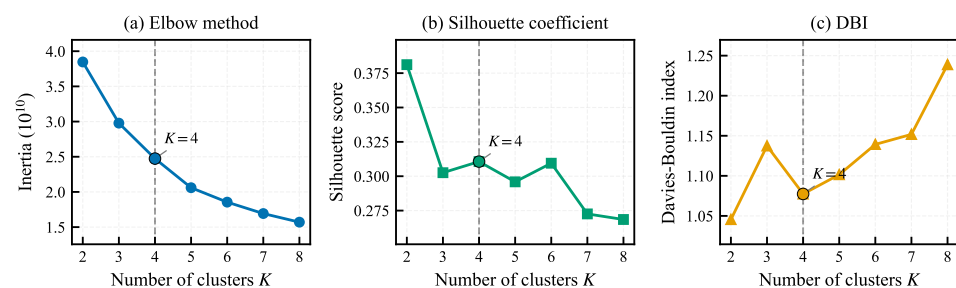


Figure 2. Evaluation of the selected number of clusters K .

Table 4. Sensitivity of the selected BESS configuration to the number of typical scenarios.

K	$\alpha = 0.05$		$\alpha = 0.15$	
	Selected $P_B/E_B/h$ (MW/MWh/h)	NPV (10^9 CNY)	Selected $P_B/E_B/h$ (MW/MWh/h)	NPV (10^9 CNY)
2	300/300/1	54.3695	1000/1000/1	53.1153
3	300/300/1	56.3470	1000/1000/1	55.0878
4	300/300/1	54.9418	1000/1000/1	53.6806
5	300/300/1	54.8640	1000/1000/1	53.7207
6	300/300/1	54.1845	1000/1000/1	53.0265

After fixing $K = 4$, each centroid is mapped to the closest historical day rather than used directly as a synthetic profile, thereby preserving the original intra-day ramps and peak–valley variations. The resulting four typical scenarios are weighted by their cluster sizes and used in the expected-NPV evaluation, whereas the three zero-probability extreme scenarios in Table 5 are used only for stress testing.

Table 5. Typical scenarios and extreme stress test scenarios.

Scenario	Type	Weight	Original Date	RES Energy (MWh/day)
Typical 1	Typical	0.315	4 September 2023	52,063.48
Typical 2	Typical	0.279	20 September 2023	98,544.06
Typical 3	Typical	0.271	26 October 2023	23,015.44
Typical 4	Typical	0.134	4 December 2023	61,291.13
Extreme A	Stress test	0	27 March 2023	56,757.70
Extreme B	Stress test	0	25 January 2023	4,852.80
Extreme C	Stress test	0	13 March 2023	121,396.31

Note: Typical-scenario weights sum to 1 before rounding. Stress test scenarios are assigned zero probability and are used only for boundary condition analysis.

4.2. Comparison Between Centroid-Based and Real Day-Mapped Scenarios

To further examine the effect of scenario construction, an additional comparison is conducted between centroid-based typical scenarios and real day-mapped typical scenarios. The two methods use the same K-means clustering results, cluster weights, candidate BESS configurations, economic parameters, reserve requirements, and PCC limits. The only difference is that the centroid-based method directly uses the K-means cluster centroids as synthetic typical days, whereas the real day-mapped method selects the real historical day closest to each centroid.

Figure 3 compares the centroid-based synthetic profiles with the real day-mapped profiles for the four clusters, with panels (a)–(d) corresponding to Clusters 1–4, respectively. The centroid-based profiles are generally smoother because they are obtained by averaging the daily samples within each cluster, whereas the real day-mapped profiles retain more pronounced local peak–valley variations and short-term ramps. The weighted profile characteristics of the two scenario construction methods are further summarized in Table 6. Compared with the centroid-based profiles, the real day-mapped profiles increase the weighted ramp sum from 5564.01 MW to 8138.46 MW, corresponding to an increase of 46.27%. The weighted maximum ramp also increases from 171.57 MW to 306.40 MW, representing an increase of 78.59%. These results indicate that real-day mapping preserves stronger intra-day fluctuation characteristics than the direct use of clustering centroids.

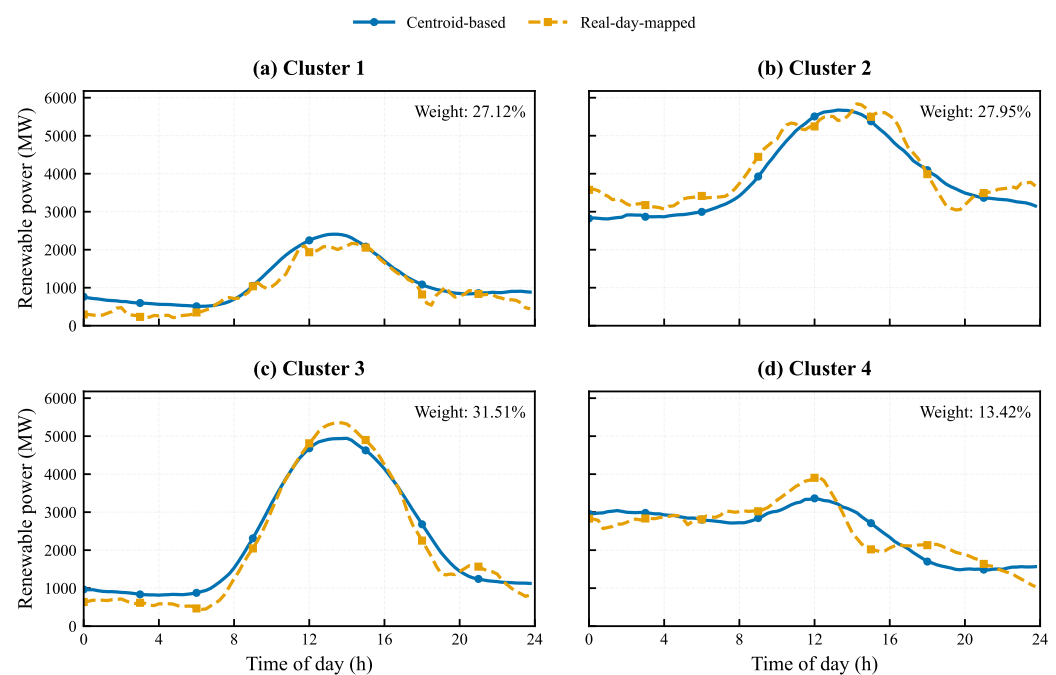


Figure 3. Comparison between centroid-based synthetic typical scenarios and real day-mapped typical scenarios.

Table 6. Comparison of profile characteristics between centroid-based and real day-mapped scenarios.

Metric	Centroid-Based	Real Day-Mapped	Difference	Relative Change
Daily energy (MWh)	59,016.81	58,412.57	−604.24	−1.02%
Std power (MW)	1003.39	1053.51	50.12	4.99%
Ramp sum (MW)	5564.01	8138.46	2574.45	46.27%
Max. ramp (MW)	171.57	306.40	134.84	78.59%
Max. power (MW)	4248.33	4431.43	183.10	4.31%

The corresponding BESS sizing results are reported in Table 7. In terms of selected capacity, the two scenario construction methods lead to the same BESS configuration under the tested candidate set. When $\alpha = 0.05$, both methods select 300 MW/300 MWh with a duration of 1 h. When $\alpha = 0.15$, both methods select 1000 MW/1000 MWh with a duration of 1 h. However, the economic evaluation results are not identical. The NPV obtained using real day-mapped scenarios is approximately 6.21×10^8 CNY lower than that obtained using centroid-based scenarios under both reserve ratio settings, corresponding to a relative difference of about 1.1%. This difference is mainly caused by the slightly lower weighted renewable energy and energy sale volume of the selected real historical days.

Table 7. Comparison of selected BESS configurations under different scenario construction methods.

α	Method	Config.	NPV	Deg. Cost	EFC
0.05	Centroid	300/300/1	555.63	0.2621	1.1938
0.05	Real day	300/300/1	549.42	0.2612	1.1884
0.15	Centroid	1000/1000/1	543.02	0.9169	1.2344
0.15	Real day	1000/1000/1	536.81	0.9139	1.2296

Note: “Config.” denotes the selected BESS power, energy capacity, and duration in MW/MWh/h. NPV is reported in 10^8 CNY. “Deg. cost” denotes the annual degradation cost in 10^8 CNY/year. “Centroid” denotes centroid-based synthetic scenarios, and “Real day” denotes real day-mapped scenarios.

Moreover, under $\alpha = 0.15$, the real day-mapped scenarios lead to higher reserve shortages for undersized BESS configurations. For example, when $P_B = 300, 500$, and 800 MW, the reserve shortages under real day-mapped scenarios are higher than those under centroid-based scenarios by 386.93, 384.64, and 10.30 MWh-eq/day, respectively. This indicates that centroid-based synthetic profiles may underestimate reserve pressure in insufficient-capacity cases. Therefore, although real-day mapping does not change the selected BESS configuration in this experiment, it provides a more realistic input basis for evaluating intra-day fluctuation, reserve shortage, and degradation-related indicators.

4.3. Baseline Capacity Search Under the Reference Case

The reference case is evaluated with $\alpha^{\text{up}} = \alpha^{\text{down}} = 0.05$ and $P_{\text{max}}^{\text{PCC, sell}} = P_{\text{max}}^{\text{PCC, buy}} = 10,000$ MW. The candidate set contains five rated power levels, $P_B = \{300, 500, 800, 1000, 1500\}$ MW, and three storage durations, $\{1, 2, 4\}$ h. This gives 15 candidate BESS configurations in total. For each configuration, the lower-level dispatch problem is solved in the four typical scenarios, and the scenario-weighted operating profit, degradation cost, O&M cost, and net present value (NPV) are calculated.

Figure 4 presents the NPV distribution over the candidate power duration combinations, and Table 8 lists selected representative configurations from the full search. Under the reference setting, the highest NPV within the predefined candidate set is obtained by the 300 MW/300 MWh configuration, corresponding to a storage duration of 1 h. This configuration yields an NPV of 54.942 billion CNY, with an annual operating profit of 8.351 billion CNY and an annual battery degradation cost of 0.026 billion CNY.

The reference case provides a benchmark for the subsequent comparisons. With a 5% reserve ratio and non-binding PCC limits, the economic benefit of increasing BESS capacity is not sufficient to offset the additional investment and degradation costs. As a result, the smallest 1 h configuration is selected within the predefined candidate set. The following subsections examine how this result changes when the scenario construction method, reserve requirement, PCC headroom, and reserve deliverability modeling assumptions are varied.

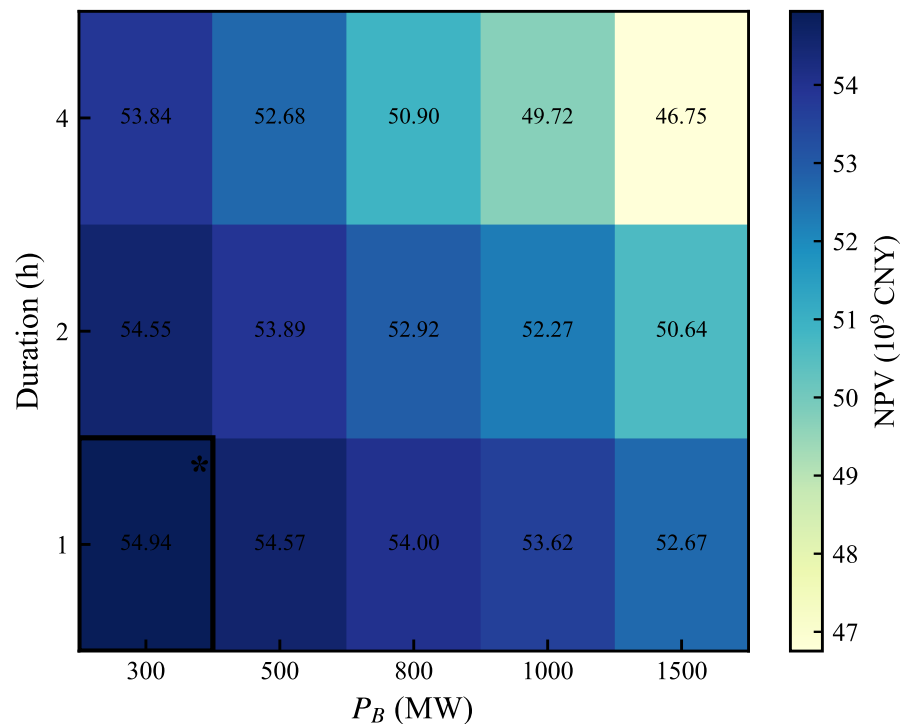


Figure 4. NPV distribution of the baseline capacity search. The asterisk marks the maximum NPV, and the black square frame highlights the corresponding best-performing configuration (300 MW/300 MWh/1 h) within the tested candidate set.

Table 8. Representative results of the baseline capacity search.

Rank	P_B (MW)	E_B (MWh)	Duration (h)	NPV (10^8 CNY)	Operation Profit (10^8 CNY/Year)	Degradation Cost (10^8 CNY/Year)
1	300	300	1	549.42	83.51	0.26
2	500	500	1	545.66	84.13	0.53
3	300	600	2	545.53	83.89	0.61
4	800	800	1	539.95	85.05	0.93
7	1000	1000	1	536.17	85.67	1.20
15	1500	6000	4	467.54	93.13	7.48

Note: The table reports selected configurations from the full 15-combination search. NPV includes investment cost, annual operating profit, O&M cost, and post-processed rainflow degradation cost.

4.4. Benchmark and Ablation Comparison

To distinguish the effects of key modeling assumptions, six benchmark and ablation formulations, denoted as B0–B5, are evaluated under $\alpha^{\text{up}} = \alpha^{\text{down}} = 0.15$. This reserve level provides clearer differentiation than the reference value of 0.05 while remaining satisfiable within the predefined candidate set. B0 represents energy-oriented sizing, B1 adds optional reserve revenue without prescribed reserve responsibility, and B2 imposes prescribed reserve requirements using only BESS power headroom. B3 further incorporates dual-endpoint SOC margins and PCC headroom but excludes degradation cost from the candidate-level NPV. B4 is the proposed formulation with post-dispatch rainflow degradation evaluation, whereas B5 introduces a throughput penalty into the dispatch objective as a degradation-aware operational benchmark. Thus, B0, B1, and B5 are literature-informed benchmarks, while B2 and B3 are controlled ablation variants. Their modeling components are summarized in Table 9. Table 10 summarizes the resulting capacity selections, operating metrics, and rainflow-based degradation indicators.

Table 9. Modeling components of the benchmark and ablation formulations.

Model	Energy Trading	Reserve Revenue	Prescribed Reserve	Power Headroom	Dual-Endpoint SOC Margin	PCC Headroom	Rainflow Cost In NPV	Throughput Penalty In Dispatch
B0 [5,12]	✓	–	–	–	–	–	–	–
B1 [20,27]	✓	✓	–	✓	✓	✓	–	–
B2	✓	✓	✓	✓	–	–	–	–
B3	✓	✓	✓	✓	✓	✓	–	–
B4	✓	✓	✓	✓	✓	✓	✓	–
B5 [29,33]	✓	✓	✓	✓	✓	✓	✓	✓

Note: A check mark (✓) indicates that the corresponding modeling component is included in the formulation, whereas an en dash (–) indicates that it is not included.

Table 10. Benchmark and ablation results under the moderate-reserve requirement.

Model	Best P_B/E_B (MW/MWh)	Duration (h)	NPV (Billion CNY)	Daily Throughput (MWh/day)	EFC (Cycle/day)	Annual Rainflow Degradation (Million CNY/Year)	Reserve Shortage (MWh-eq/Day)
B0	300/300	1	54.859	961.3	1.6000	40.59	N/A
B1	300/300	1	55.181	961.3	1.6000	40.59	N/A
B2	1000/1000	1	54.510	3204.2	1.6000	135.31	0
B3	1000/1000	1	54.294	2462.4	1.2296	91.39	0
B4	1000/1000	1	53.681	2462.4	1.2296	91.39	0
B5	1000/1000	1	53.731	1328.3	0.6633	47.34	0

B0 and B1 select the minimum tested configuration of 300 MW/300 MWh, whereas B2–B5 select 1000 MW/1000 MWh. The comparison between B1 and B2 therefore shows that the increase in selected rated power is driven primarily by the prescribed reserve responsibility rather than by optional reserve revenue alone. Accordingly, representing reserve only as an optional revenue stream may underestimate the capacity required for prescribed grid-support duties.

Adding the dual-endpoint SOC and PCC headroom constraints in B3 reduces the weighted daily battery throughput from 3204.2 to 2462.4 MWh, the EFC from 1.6000 to 1.2296 cycle/day, and the diagnostic annual rainflow degradation cost from CNY 135.31 million to CNY 91.39 million while retaining the same selected configuration as B2. B3 and B4 also produce identical dispatch trajectories because the rainflow cost in B4 is evaluated only after dispatch. Its inclusion in the candidate-level economic evaluation reduces the NPV from CNY 54.294 billion to CNY 53.681 billion without changing the lower-level operating decisions.

In contrast, the throughput penalty in B5 directly influences dispatch and reduces the weighted daily throughput from 2462.44 to 1328.30 MWh, the EFC from 1.2296 to 0.6633 cycle/day, and the annual rainflow degradation cost from CNY 91.39 million to CNY 47.34 million. Nevertheless, B4 and B5 select the same 1000 MW/1000 MWh configuration, with NPVs of CNY 53.681 billion and CNY 53.731 billion, respectively. These results show that degradation-aware dispatch materially changes cycling behavior but does not alter the selected capacity within the tested candidate set. Overall, the contribution of B4 lies in integrating prescribed bidirectional reserve requirements, power and energy deliverability constraints, PCC headroom, chronology-preserving scenarios, and post-dispatch rainflow degradation evaluation within one capacity sizing framework.

4.5. Sensitivity to Operational Modeling Assumptions

The baseline reserve power formulation represents upward reserve through additional discharging and downward reserve through additional charging, without explicitly crediting reserve delivered by reducing the opposite scheduled action. To examine whether this conservative representation affects capacity selection, it is compared with the flexible

net power modulation formulation in Section 3.6 under $\alpha^{\text{up}} = \alpha^{\text{down}} = 0.15$ and 0.25. For each reserve level and formulation, all 15 candidate configurations are evaluated in the four typical scenarios, resulting in 240 MILP instances.

As shown in Table 11, both formulations select 1000 MW/1000 MWh/1 h at $\alpha = 0.15$ and 1500 MW/1500 MWh/1 h at $\alpha = 0.25$, with zero probability-weighted reserve shortage. The flexible formulation increases the corresponding NPVs by CNY 42.84 million and CNY 63.04 million because it recognizes additional modulation from reducing scheduled charging or discharging. Thus, the baseline formulation provides a more conservative operational valuation, but no detectable capacity oversizing is observed within the predefined candidate set and tested reserve levels. This result does not imply equivalence under arbitrary operating conditions or candidate resolutions.

The baseline 96-period model contains 1345 variables, including 384 binaries, and approximately 1924 constraints, whereas the flexible formulation contains 2113 variables, including 768 binaries, and approximately 2884 constraints. All 240 instances are solved to optimality without reaching the time limit. Because the flexible formulation increases model complexity while leaving the selected configurations unchanged, the baseline formulation is retained for the main sizing analyses.

Table 11. Comparison of the baseline and flexible reserve power formulations.

Reserve Ratio	Reserve Power Formulation	Best P_B/E_B (MW/MWh)	Duration (h)	NPV (10^9 CNY)	NPV Gain (10^6 CNY)	Reserve Shortage (MWh-eq/Day)
0.15	Baseline	1000/1000	1	53.68060	–	0
0.15	Flexible	1000/1000	1	53.72344	42.84	0
0.25	Baseline	1500/1500	1	52.79333	–	0
0.25	Flexible	1500/1500	1	52.85637	63.04	0

The daily cyclic terminal-SOC assumption is further tested using four 48 h windows, each linking a real day-mapped typical scenario to its actual following day. The daily cyclic formulation restores the initial SOC at both the 24 h boundary and the end of the 48 h horizon, whereas the linked formulation removes the intermediate reset while retaining energy neutrality at 48 h. As shown in Table 12, both treatments select 1000 MW/1000 MWh/1 h in the reserve-driven case and 300 MW/300 MWh/1 h in the PCC-constrained case. The results are identical in the reserve-driven case; in the PCC-constrained case, linking the two days increases the NPV by only CNY 0.727 million and changes the curtailment and EFC only marginally. Therefore, the daily cyclic condition does not materially affect capacity selection within the tested 48 h windows, although longer-term chronological energy shifting remains outside the scope of this benchmark.

Table 12. Effect of terminal-SOC treatment within identical 48 h operating windows.

Case	Terminal-SOC Treatment	Configuration	NPV	Curtailment	EFC
R-48 h	Daily cyclic	1000/1000/1	44.323685	0.000	1.299878
R-48 h	Linked 48 h	1000/1000/1	44.323685	0.000	1.299878
P-48 h	Daily cyclic	300/300/1	32.464532	3922.346	1.429714
P-48 h	Linked 48 h	300/300/1	32.465259	3922.425	1.422740

Note: Case R-48 h denotes the reserve-driven 48 h benchmark with $\alpha = 0.15$ and PCC export/import limits of 10,000/10,000 MW, whereas Case P-48h denotes the PCC-constrained 48 h benchmark with $\alpha = 0.05$ and PCC export/import limits of 4000/10,000 MW. Each benchmark uses four linked two-day windows rather than the probability-weighted typical-day evaluation. Configurations are reported in MW/MWh/h, NPV in billion CNY, curtailment in MWh/day, and EFC in cycle/day.

4.6. Sensitivity to the Reserve Requirement Level

The reserve ratio sensitivity analysis examines how the prescribed active support responsibility affects BESS capacity selection. The upward and downward reserve ratios

are set to be equal, $\alpha^{\text{up}} = \alpha^{\text{down}} = \alpha$, and $\alpha \in \{0.05, 0.15, 0.25, 0.35\}$ is evaluated using the same 15 candidate configurations, real day-mapped typical scenarios, PCC limits, reserve duration, and economic and technical parameters.

As shown in Table 13, increasing α from 0.05 to 0.15 and 0.25 changes the best-performing configuration within the predefined candidate set from 300 MW/300 MWh/1 h to 1000 MW/1000 MWh/1 h and 1500 MW/1500 MWh/1 h, respectively, while maintaining zero probability-weighted reserve shortage. The corresponding NPV decreases from CNY 54.942 billion to CNY 53.681 billion and CNY 52.793 billion because the additional reserve value does not fully offset the increased investment cost. At $\alpha = 0.35$, the largest original 1 h candidate, 1500 MW/1500 MWh, is selected but retains a shortage of 2355.75 MWh-eq/day, indicating that the original power rating range is insufficient for this reserve level.

Table 13. Sensitivity of the selected BESS configuration to the reserve requirement level.

Reserve Ratio α	Best P_B/E_B (MW/MWh)	Duration (h)	NPV (10^9 CNY)	Reserve Shortage (MWh-eq/Day)	Weighted EFC (cycle/Day)
0.05	300/300	1	54.942	0.00	1.1884
0.15	1000/1000	1	53.681	0.00	1.2296
0.25	1500/1500	1	52.793	0.00	1.1880
0.35	1500/1500	1	23.976	2355.75	0.9160

The weighted EFC varies non-monotonically from 1.1884 to 1.2296, 1.1880, and 0.9160 cycle/day as α increases. This behavior reflects both the changing SOC trajectory and its normalization by the selected energy capacity. At $\alpha = 0.35$, the reserve shortage also limits the reserve actually delivered by the BESS.

To examine dependence on candidate grid resolution, the rated power is refined from 100 to 2500 MW in 100 MW increments and the duration set is expanded to $\{0.5, 1.0, 1.5, 2.0, 3.0, 4.0\}$ h. For $\alpha = 0.35$, the upper power bound is further extended to 3500 MW. Table 14 shows that the best-performing rated powers become 300, 1100, 1800, and 2600 MW as α increases from 0.05 to 0.35, with the higher-reserve cases favoring a duration of 0.5 h. At $\alpha = 0.25$, the best-performing 1800 MW/900 MWh configuration retains a small shortage of 5.003 MWh-eq/day, whereas the highest-NPV zero-shortage alternative is 1900 MW/950 MWh, with an NPV reduction of CNY 76.5 million. At $\alpha = 0.35$, the extended grid selects 2600 MW/1300 MWh and eliminates the shortage observed under the original grid. Thus, the exact configuration depends on the bounds and resolution of the candidate set but the trend of increasing preferred rated power with stronger reserve responsibility remains stable.

Table 14. Comparison of BESS sizing results under the original and refined candidate grids.

α	Original Candidate Grid			Refined Candidate Grid			NPV Gain (10^6 CNY)
	Selected $P_B/E_B/h$	NPV (10^9 CNY)	Shortage (MWh-eq/Day)	Selected $P_B/E_B/h$	NPV (10^9 CNY)	Shortage (MWh-eq/Day)	
0.05	300/300/1.0	54.9418	0.0000	300/300/1.0	54.9418	0.0000	0.0
0.15	1000/1000/1.0	53.6806	0.0000	1100/550/0.5	53.9666	0.0000	286.0
0.25	1500/1500/1.0	52.7933	0.0000	1800/900/0.5	52.9429	5.0030	149.6
0.35	1500/1500/1.0	23.9758	2355.7500	2600/1300/0.5	51.9042	0.0000	27,928.4

Note: BESS configurations are reported as rated power/energy capacity/duration in MW/MWh/h. The refined grid includes power ratings from 100 to 2500 MW in 100 MW increments and durations of 0.5, 1.0, 1.5, 2.0, 3.0, and 4.0 h. For $\alpha = 0.35$, the power range is further extended to 3500 MW because the initial refined-grid solution reaches the 2500 MW boundary. The NPV gain is calculated relative to the original-grid result under the same reserve ratio.

4.7. Impact of PCC Channel Constraints on the Functional Role of BESS

The PCC sensitivity analysis evaluates how the export capability of the grid connection influences the selected BESS configuration. In this test, the PCC import limit is kept unchanged, while the PCC export limit is gradually tightened. Table 15 reports the selected configurations, NPVs, and weighted renewable curtailment under different PCC export limits.

Table 15. Sensitivity analysis of the PCC export limit.

PCC Limit (MW)	Best P_B/E_B (MW/MWh)	Duration (h)	NPV (10^8 CNY)	Curtailment (MWh/Day)
10,000	300/300	1	549.42	0.00
6000	300/300	1	548.92	15.15
5000	500/1000	2	494.39	1326.60
4000	1000/4000	4	364.03	3691.45

When the PCC export limit is 10,000 MW, renewable curtailment does not occur and the selected configuration is 300 MW/300 MWh. Tightening the export limit to 6000 MW causes only a small amount of curtailment, and the selected configuration remains unchanged. In contrast, when the export limit is reduced to 5000 MW and 4000 MW, the selected configurations increase to 500 MW/1000 MWh and 1000 MW/4000 MWh, respectively. The corresponding weighted curtailment also increases to 1326.60 and 3691.45 MWh/day.

These results show that the PCC export limit affects not only the economic value of storage, but also the type of storage capacity that becomes valuable. With a loose PCC channel, the selected BESS is mainly a short-duration resource for reserve provision and limited energy shifting. As the PCC export limit becomes binding, more renewable energy must be absorbed locally and shifted to later periods. In this situation, longer-duration storage becomes more attractive because it can reduce curtailment and provide intertemporal energy transfer. This result complements the reserve ratio sensitivity analysis: under the base economic and cycling-only degradation assumptions, higher reserve requirements primarily increase the selected rated power, whereas tighter PCC export limits increase the value of energy capacity and storage duration.

4.8. Sensitivity to Economic, Reserve Service, and Degradation Parameters

A one-at-a-time sensitivity analysis is conducted for the principal economic, reserve service, and degradation-related assumptions, including the power-related investment cost c_P , energy-related investment cost c_E , electricity price level, reserve price, reserve shortage penalty ρ^{short} , curtailment penalty ρ^{curt} , and reserve deployment duration τ . Unless otherwise specified, the analysis uses $\alpha^{\text{up}} = \alpha^{\text{down}} = 0.15$, PCC export/import limits of 10000/10000 MW, the four real day-mapped typical scenarios, and the original 15 candidate configurations. For each parameter level, all candidate configurations are re-evaluated and the highest-NPV configuration within the predefined candidate set is identified. The tests requiring dispatch reoptimization involve 660 MILP instances, all of which are solved to optimality with a maximum MIP gap of 8.11×10^{-5} . The electricity price multiplier is applied uniformly to the valley, flat, and peak prices, whereas the reserve price multiplier is applied to both λ^{up} and λ^{down} . The results are summarized in Table 16.

Among the tested parameters, the electricity price level has the largest effect on life-cycle economic performance. Increasing its multiplier from 0.8 to 1.2 changes the selected NPV from CNY 42.4291 billion to CNY 64.9319 billion, although the selected configuration remains 1000 MW/1000 MWh/1 h. Variations in c_P , c_E , and the reserve price also leave

the selected configuration unchanged within the tested ranges. Nevertheless, c_E directly affects the rainflow degradation cost valuation: the annual degradation cost increases from CNY 73.11 million to CNY 109.66 million as c_E rises from 0.8 to 1.2 times its baseline value.

Table 16. Sensitivity to economic and reserve service parameters under the moderate-reserve condition.

Parameter	Tested Levels	Selected $P_B/E_B/h$ (MW/MWh/h)	Selected NPV Range (10^9 CNY)	Maximum Shortage (MWh-eq/Day)
Power investment cost c_P	1.20/1.50/1.80 $\times 10^6$ CNY/MW	1000/1000/1	53.3403–54.0208	0.00
Energy investment cost c_E	0.96/1.20/1.44 $\times 10^6$ CNY/MWh	1000/1000/1	53.2857–54.0754	0.00
Electricity price multiplier	0.8/1.0/1.2	1000/1000/1	42.4291–64.9319	0.00
Reserve price $\lambda^{\text{up/down}}$	5/10/15 CNY/(MW·h)	1000/1000/1	53.1310–54.2301	0.00
Shortage penalty ρ^{short}	2500/5000/10,000 CNY/(MW·h)	1000/1000/1	53.6805–53.6806	0.00
Curtailement penalty ρ^{curt}	500/1000/2000 CNY/MWh	1000/1000/1	53.6805–53.6805	0.00
Reserve duration τ	0.25/0.50/1.00 h	1000/1000/1; 1500/1500/1; 1000/2000/2	51.2663–53.6806	83.60

Note: The selected configuration is listed in the same order as the tested levels when it changes across the parameter range. The NPV range reports the minimum and maximum selected NPVs among the tested levels.

The reserve deployment duration is the only parameter in Table 16 that changes the selected configuration. Increasing τ from 0.25 to 0.50 h changes the selection from 1000 MW/1000 MWh/1 h to 1500 MW/1500 MWh/1 h. At $\tau = 1.00$ h, the selected configuration becomes 1000 MW/2000 MWh/2 h and retains a probability-weighted reserve shortage of 83.60 MWh-eq/day. Thus, a longer sustainment requirement increases the importance of SOC energy margins and may favor greater energy capacity or storage duration.

To examine the potential bias caused by the simplified cycling-only degradation model, planning-level calendar aging and C-rate corrections are applied to the same B4 dispatch trajectories. The assumed calendar life is varied from 20 to 10 years, while the C-rate correction increases the rainflow cycling loss according to the scenario-level 95th-percentile battery-side C-rate. These corrections are illustrative sensitivity assumptions rather than chemistry-specific aging models. As summarized in Table 17, all tested settings retain the 1000 MW/1000 MWh/1 h configuration in the reserve-driven case, although the NPV decreases from CNY 53.681 billion to CNY 52.458 billion under the combined conservative setting. In the PCC-constrained case, a 20-year calendar life assumption already changes the selected configuration from 1000 MW/4000 MWh/4 h to 300 MW/300 MWh/1 h, with a rank 1–rank 2 NPV gap of CNY 102.559 million. The C-rate correction alone changes the selection only under the conservative setting, for which 800 MW/3200 MWh/4 h exceeds the original 1000 MW/4000 MWh/4 h candidate by CNY 9.479 million. Therefore, neglecting calendar aging and the C-rate may cause not only NPV overestimation but also capacity ranking bias under energy capacity-intensive operating conditions, while the reserve-driven sizing result remains stable within the tested assumptions.

Under the moderate-reserve base condition adopted in this subsection, the reserve shortage and curtailment penalties have negligible effects because the selected configuration produces neither reserve shortage nor renewable curtailment. Two additional conditional tests are therefore conducted under operating conditions in which the corresponding penalty terms are active. The results are reported in Table 18.

Table 17. Sensitivity of BESS sizing results to calendar aging and C-rate corrections.

Setting	Res. Config.	Res. NPV	PCC Config.	PCC NPV
Base	1000/1000/1	53.681	1000/4000/4	36.403
Calendar moderate	1000/1000/1	53.144	300/300/1	35.936
C-rate moderate	1000/1000/1	53.472	1000/4000/4	36.316
Combined conservative	1000/1000/1	52.458	300/300/1	35.737

Note: Res. = reserve-driven; PCC = PCC-constrained. The reserve-driven case uses $\alpha = 0.15$ and PCC export/import limits of 10000/10000 MW, whereas the PCC-constrained case uses $\alpha = 0.05$ and PCC export/import limits of 4000/10000 MW. Configurations are reported in MW/MWh/h, and NPVs are reported in billion CNY.

Table 18. Conditional sensitivity to activated reserve shortage and curtailment penalties.

Activated Condition	Penalty Level	Selected $P_B/E_B/h$ (MW/MWh/h)	NPV (10^9 CNY)	Shortage (MWh-eq/day)	Curtailment (MWh/day)
$\tau = 1.00$ h, $\alpha = 0.15$	$\rho^{\text{short}} = 2500$	1000/2000/2	51.7782	83.60	0.00
$\tau = 1.00$ h, $\alpha = 0.15$	$\rho^{\text{short}} = 5000$	1000/2000/2	51.2663	83.60	0.00
$\tau = 1.00$ h, $\alpha = 0.15$	$\rho^{\text{short}} = 10,000$	1500/3000/2	50.6786	0.00	0.00
$P_{\text{max}}^{\text{PCC, sell}} = 5000$ MW, $\alpha = 0.05$	$\rho^{\text{curt}} = 500$	300/300/1	51.4312	0.00	1681.94
$P_{\text{max}}^{\text{PCC, sell}} = 5000$ MW, $\alpha = 0.05$	$\rho^{\text{curt}} = 1000$	500/1000/2	49.4386	0.00	1326.60
$P_{\text{max}}^{\text{PCC, sell}} = 5000$ MW, $\alpha = 0.05$	$\rho^{\text{curt}} = 2000$	1000/4000/4	47.1292	0.00	411.66

Under $\tau = 1.00$ h and $\alpha = 0.15$, increasing ρ^{short} from 5000 to 10000 CNY/(MW·h) changes the selected configuration from 1000 MW/2000 MWh/2 h to 1500 MW/3000 MWh/2 h and eliminates the probability-weighted shortage of 83.60 MWh-eq/day. Under the 5000-MW PCC export limit, increasing ρ^{curt} from 500 to 2000 CNY/MWh changes the selected configuration from 300 MW/300 MWh/1 h to 1000 MW/4000 MWh/4 h and reduces renewable curtailment from 1681.94 to 411.66 MWh/day, a reduction of approximately 75.5%. These results show that the two penalty coefficients affect capacity selection when their corresponding shortage or curtailment variables are active.

A hard-reserve benchmark is further evaluated by fixing the upward and downward reserve shortage variables to zero in all typical scenarios and dispatch periods. The comparison is conducted at $\alpha^{\text{up}} = \alpha^{\text{down}} = 0.15$ under reserve deployment durations of $\tau = 0.25$ and 1.00 h while all other settings remain unchanged. The comparison results are summarized in Table 19.

Table 19. Comparison between the soft-shortage and hard-reserve formulations.

Reserve Duration (h)	Formulation	Feasible Candidates	Selected $P_B/E_B/h$ (MW/MWh/h)	NPV (10^9 CNY)	Reserve Shortage (MWh-eq/day)	NPV Difference (10^6 CNY)
0.25	Soft-shortage	15	1000/1000/1	53.6806	0.0000	–
0.25	Hard-reserve	6	1000/1000/1	53.6806	0.0000	0.0
1.00	Soft-shortage	15	1000/2000/2	51.2663	83.6044	–
1.00	Hard-reserve	3	1500/3000/2	50.6786	0.0000	–587.7

At $\tau = 0.25$ h, the soft-shortage and hard-reserve formulations both select 1000 MW/1000 MWh/1 h because the soft-constrained solution already maintains zero reserve shortage. At $\tau = 1.00$ h, strict enforcement changes the selected configuration from 1000 MW/2000 MWh/2 h to 1500 MW/3000 MWh/2 h, eliminates the shortage of 83.6044 MWh-eq/day, and reduces the NPV by approximately CNY 587.7 million. Thus, the hard-reserve formulation affects capacity selection when the highest-NPV soft-constrained configuration retains a non-zero shortage.

Overall, electricity prices primarily affect the estimated economic value of the BESS, whereas reserve deployment duration, active penalty terms, and strict zero-shortage en-

forcement can alter the balance between rated power and energy capacity. The degradation sensitivities further show that omitted calendar aging and C-rate effects may influence candidate rankings, particularly under energy capacity-intensive PCC-constrained operation.

4.9. Rainflow Degradation and Extended Stress Testing

Battery degradation is evaluated by applying rainflow counting to the optimized SOC trajectories. For the reference case, including the post-processed degradation cost does not change the selected configuration, which remains 300 MW/300 MWh. However, the NPV of this configuration decreases from 55.117 billion CNY to 54.942 billion CNY, corresponding to a reduction of 0.175 billion CNY. In this case, the rainflow-based degradation assessment mainly serves as an economic correction to the life-cycle evaluation. It does not alter the selected capacity, but it reduces the risk of overestimating the economic performance of configurations that experience frequent or deep cycling. The degradation-aware benchmark in Section 4.4 further shows that allowing a throughput-based cycling surrogate to influence dispatch substantially reduces battery throughput, EFC, and rainflow degradation cost, although the selected configuration remains unchanged within the predefined candidate set.

Figure 5 separately presents the SOC trajectories of the selected configuration in (a) the four typical scenarios and (b) the three extreme stress test scenarios. In all tested scenarios, the SOC remains within the prescribed operating range, confirming that the optimized dispatch satisfies the battery energy constraints. As shown in panel (a), the different typical renewable generation profiles result in distinct daily charging and discharging patterns. Panel (b) further shows that the extreme scenarios produce more pronounced differences in the depth and temporal structure of SOC cycling. These scenario-dependent operating trajectories explain why the rainflow-based degradation indicators differ even for the same BESS configuration.

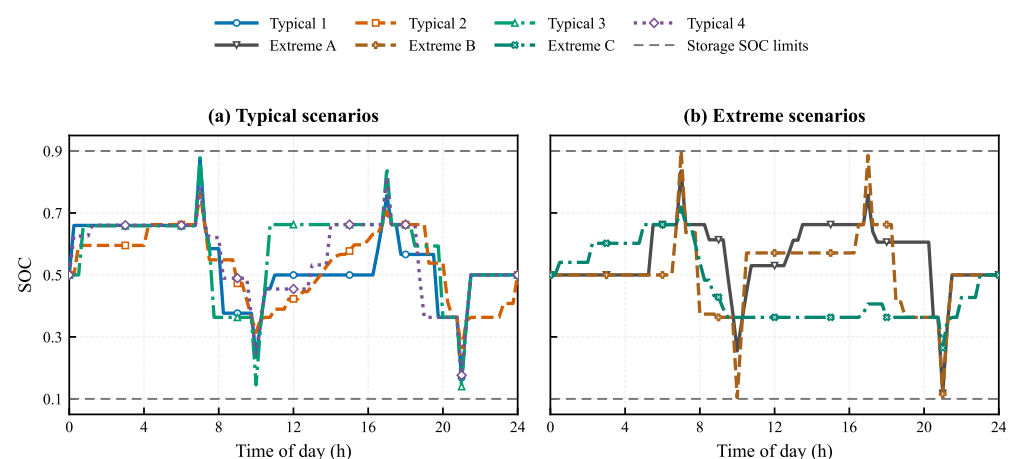


Figure 5. SOC trajectories of the selected configuration in typical and extreme scenarios.

Scenario-level EFC, degradation cost, and reserve shortage are further reported in Figure 6 and Table 20. In the four typical scenarios, the selected 300 MW/300 MWh BESS produces no renewable curtailment, no grid purchase, and no reserve shortage. This indicates that the selected configuration can satisfy the 5% reserve requirement under the tested typical operating conditions.

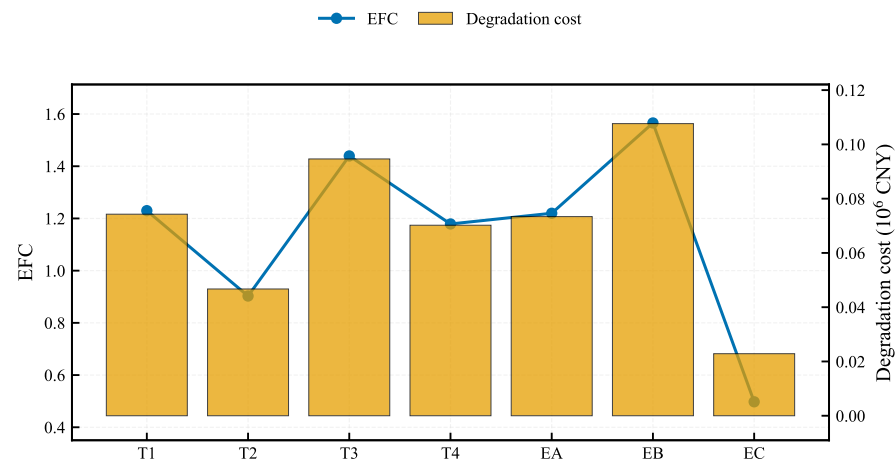


Figure 6. EFC and degradation cost in different scenarios.

Table 20. Performance of the selected configuration in typical and extreme scenarios.

Scenario	Type	Objective (10 ⁶ CNY/Day)	Degradation (10 ⁶ CNY/Day)	EFC (-)	Reserve Shortage (MWh-eq/Day)
Typical 1	Typical	20.79	0.074	1.230	0.00
Typical 2	Typical	37.97	0.047	0.903	0.00
Typical 3	Typical	9.40	0.095	1.439	0.00
Typical 4	Typical	23.61	0.070	1.179	0.00
Extreme A	Stress test	22.09	0.073	1.220	0.00
Extreme B	Stress test	2.23	0.108	1.566	0.00
Extreme C	Stress test	43.18	0.023	0.498	571.91

The stress test results show that degradation stress is not determined only by the total amount of renewable generation. Extreme B, which corresponds to the minimum-output day, has a renewable generation of only 4852.80 MWh/day, but it produces the highest EFC of 1.566 and the highest degradation cost of 0.108 million CNY/day. This result indicates that the SOC cycling structure under low-output conditions can be more severe than that under some higher-output scenarios. By contrast, Extreme C has the highest renewable generation, 121,396.31 MWh/day, but its EFC is only 0.498. Therefore, under the tested settings, battery aging is more closely associated with the rainflow cycle depth distribution of the optimized SOC trajectory than with renewable generation volume alone.

Extreme C also produces a reserve shortage of 571.91 MWh-eq/day. This result suggests that a configuration selected using only typical scenarios may still encounter reserve insufficiency under extreme high-output conditions. Although the stress test scenarios are excluded from the expected NPV calculation, they provide additional information on boundary condition risks that may not be visible from typical-scenario evaluation alone.

To further examine operational feasibility beyond the three single-day extreme scenarios, four additional stress conditions are considered: the consecutive 48 h periods with the lowest and highest renewable generation, the day containing the maximum single-step renewable ramp, and a synthetic two-sided forecast error envelope with $\epsilon = 0.10$. These stress scenarios are assigned zero probability and are not included in the expected-NPV calculation. As summarized in Table 21, all three selected configurations remain shortage-free and hard-reserve-feasible during the consecutive low-generation period. However, the reference and reserve-driven configurations are hard-feasible in only one of the four additional stress conditions. Under the sustained-high and maximum-ramp conditions, their minimum-investment zero-shortage recovery configurations within the original candidate grid are 500 MW/500 MWh/1 h and 1500 MW/1500 MWh/1 h, respectively. The

PCC-constrained 1000 MW/4000 MWh/4 h configuration remains hard-feasible under the historical Low48, High48, and Ramp24 conditions, but the High48 condition causes 24,308.897 MWh/day of renewable curtailment. The synthetic forecast error condition is the most restrictive, producing shortages of 17,006.478, 5470.730, and 541.467 MWh-eq/day in Cases B, R, and P, respectively. When the error envelope is increased to $\epsilon = 0.20$, no zero-shortage configuration is found within the original 15-candidate grid for any of the three cases. These results indicate that probability-weighted typical-scenario sizing does not guarantee feasibility under all chronological and uncertainty-related boundary conditions, motivating future robust, chance-constrained, or forecast error-aware sizing formulations.

Table 21. Performance of the selected BESS configurations in the extended stress scenarios.

Case	Config.	Zero-Short. Count	Worst Stress	Max Short.	Max Curtail.	Max EFC
B	300/300/1	1/4	$\epsilon = 0.10$	17,006.478	0.000	1.5501
R	1000/1000/1	1/4	$\epsilon = 0.10$	5470.730	0.000	1.5551
P	1000/4000/4	3/4	$\epsilon = 0.10$	541.467	24,308.897	1.8223

Note: Cases B, R, and P denote the baseline, reserve-driven, and PCC-constrained cases, respectively. Config. = configuration; zero-short. Count = zero-shortage feasible count; Max short. = maximum shortage; Max curtail. = maximum curtailment. Configurations are reported in MW/MWh/h, maximum reserve shortage in MWh-eq/day, maximum renewable curtailment in MWh/day, and maximum EFC in cycle/day. The zero-shortage feasible count gives the number of additional stress conditions with zero reserve shortage.

4.10. No-BESS Baseline and Incremental Value

A no-BESS baseline is constructed to separate the contribution of storage from the inherent energy value of the renewable power plant. In this baseline, the plant is evaluated under the same reserve requirement, but the storage capacity is set to zero, i.e., $P_B = E_B = 0$. Since no reserve can be provided by the BESS, insufficient reserve capability is reflected through shortage penalties.

Table 22 compares the no-BESS baseline with the selected BESS configuration. The no-BESS baseline gives an NPV of −16.052 billion CNY and a weighted reserve shortage of 5841.26 MWh-eq/day. After installing the selected 300 MW/300 MWh BESS, the reserve shortage in the typical scenarios is eliminated, and the incremental NPV reaches 70.994 billion CNY.

Table 22. Incremental value of the selected BESS configuration against the no-BESS baseline.

Case	P_B/E_B (MW/MWh)	NPV (10^8 CNY)	Reserve Shortage (MWh-eq/Day)	Incremental NPV (10^8 CNY)
No-BESS baseline	0/0	−160.52	5841.26	—
Selected BESS	300/300	549.42	0.00	709.94

Note: The no-BESS baseline is evaluated under the same reserve requirement as the BESS-equipped plant. Therefore, its negative NPV mainly results from reserve shortage penalties.

The incremental NPV should not be interpreted as pure arbitrage revenue. Because the no-BESS baseline is subject to the same prescribed reserve requirement, its negative NPV is mainly caused by reserve shortage penalties. The comparison therefore quantifies the value of satisfying the reserve requirement under the assumed shortage penalty setting rather than the standalone market profitability of the BESS. The result shows that, in reserve-constrained sizing, the value of storage comes not only from energy trading, but also from reducing reserve insufficiency and avoiding shortage penalties.

5. Conclusions

This paper develops a reserve deliverability-constrained BESS sizing framework for renewable power plants. Prescribed upward and downward reserve requirements are evaluated by jointly considering BESS power headroom, dual-endpoint SOC energy margins, and PCC export/import headroom. Real day-mapped typical scenarios and zero-probability stress scenarios are used for economic evaluation and boundary condition testing, respectively. Cycling degradation is assessed by applying rainflow counting after dispatch, and the resulting cost is included in the candidate-level economic comparison rather than in the lower-level MILP objective.

The results show that reserve requirements and PCC constraints affect different dimensions of BESS sizing. Within the original candidate set, increasing the reserve ratio from 0.05 to 0.25 changes the best-performing configuration from 300 MW/300 MWh to 1500 MW/1500 MWh. Under the base economic assumptions and the cycling-only degradation model, tightening the PCC export limit from 10,000 to 4000 MW changes the best-performing configuration from 300 MW/300 MWh/1 h to 1000 MW/4000 MWh/4 h. The reserve-driven increase in rated power remains stable across the tested sensitivity analyses, whereas the duration-related PCC result is more sensitive to calendar aging assumptions. The benchmark and ablation results further show that optional reserve revenue alone does not represent the prescribed reserve responsibility, and that ignoring SOC or PCC deliverability margins can overestimate feasible reserve capability. The selected configurations remain stable for $K = 2$ –6 and under the tested flexible reserve power and 48 h terminal-SOC formulations. However, strict zero-shortage enforcement can require additional capacity under longer reserve durations, and the stress tests show that probability-weighted typical-scenario sizing does not guarantee feasibility under all extreme or forecast error conditions.

The numerical case represents an equivalent large-scale renewable portfolio constructed from nationally aggregated Belgian wind and photovoltaic data rather than a specific single-site plant. Geographic aggregation may smooth local variability, and the reported PCC and BESS scales should therefore be interpreted at the aggregated system level. The representative-day framework cannot capture all chronological and forecast error conditions, the economic assumptions are case-specific, and the adopted rainflow model excludes detailed calendar aging, temperature, C-rate, and average-SOC effects. Accordingly, the reported configurations are the best-performing options within the tested candidate sets rather than generally applicable or globally optimal BESS sizes. Future work will focus on plant-level validation, chronological and forecast error-aware robust or chance-constrained sizing, degradation-aware dispatch models incorporating additional aging mechanisms, and the mapping of the proportional reserve representation to specific FCR, aFRR, and mFRR products by considering their activation duration requirements, bidding rules, and activation energy settlement mechanisms.

Author Contributions: Conceptualization, R.H. and X.Y.; methodology, R.H. and W.G.; software, W.G. and X.T.; validation, X.T., Z.Z., and F.Y.; formal analysis, R.H. and Z.Z.; investigation, W.G., X.T., and F.Y.; resources, F.Y. and X.Y.; data curation, Z.Z. and W.G.; writing—original draft preparation, R.H., W.G. and X.T.; writing—review and editing, Z.Z., F.Y., and X.Y.; visualization, X.T. and Z.Z.; supervision, X.Y.; project administration, X.Y. and R.H.; funding acquisition, X.Y. All authors have read and agreed to the published version of the manuscript.

Funding: This work was supported by Energy storage planning and design technologies for actively supporting the new power system (no. B704JY220064).

Data Availability Statement: The data presented in this study are available from the Elia Open Data platform <https://opendata.elia.be/>. The study uses the publicly available 2023 aggregated wind

and photovoltaic generation data for Belgium provided by Elia (Belgium's Transmission System Operator).

Conflicts of Interest: Authors Ruosong Hou, Wei Guo, Xiaolin Tan, and Ziheng Zhao were employed by State Grid Hebei Electric Power Co., Ltd. and worked at its Economic and Technological Research Institute. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. International Energy Agency. *Batteries and Secure Energy Transitions*; Technical report; International Energy Agency: Paris, France, 2024. Available online: <https://www.iea.org/reports/batteries-and-secure-energy-transitions> (accessed on 29 May 2026).
2. National Renewable Energy Laboratory. 2024 Annual Technology Baseline: Utility-Scale Battery Storage. 2024. Available online: https://atb.nrel.gov/electricity/2024/utility-scale_battery_storage (accessed on 29 May 2026).
3. Apribowo, C.H.B.; Sarjiya, S.; Hadi, S.P.; Wijaya, F.D. Optimal planning of battery energy storage systems by considering battery degradation due to ambient temperature: A review, challenges, and new perspective. *Batteries* **2022**, *8*, 290.
4. Jaradat, T.; Khatib, T. A review of battery energy storage system for renewable energy penetration in electrical power system: Environmental impact, sizing methods, market features, and policy frameworks. *Future Batter.* **2025**, *7*, 100106. <https://doi.org/10.1016/j.fub.2025.100106>.
5. Wang, B.; Wang, X.; Zhu, Z.; Wu, X. Siting and sizing of energy storage for renewable generation utilization with multi-stage dispatch under uncertainty: A tri-level model and decomposition approach. *Appl. Energy* **2023**, *344*, 121286.
6. Yuan, Z.; Zecchino, A.; Cherkaoui, R.; Paolone, M. Real-time control of battery energy storage systems to provide ancillary services considering voltage-dependent capability of DC-AC converters. *IEEE Trans. Smart Grid* **2021**, *12*, 4164–4175.
7. Lee, J.O.; Kim, Y.S. Novel battery degradation cost formulation for optimal scheduling of battery energy storage systems. *Int. J. Electr. Power Energy Syst.* **2022**, *137*, 107795.
8. Park, S.W.; Yu, J.U.; Lee, J.W.; Son, S.Y. A comprehensive review of battery-based power service applications considering degradation: Research status and model integration. *Appl. Energy* **2024**, *374*, 123879.
9. Butt, H.Z.; Li, X. Optimal planning of pv and battery resources in remote microgrids considering degradation costs: An iterative post-optimization correction-based approach. *arXiv* **2024**, arXiv:2402.01989.
10. Mantegna, G.; Ricks, W.; Manocha, A.; Patankar, N.; Mallapragada, D.; Jenkins, J. Establishing best practices for modeling multi-day energy storage in deeply decarbonized energy systems. *Environ. Res. Energy* **2024**, *1*, 045014.
11. Apa, L.; D'Alvia, L.; Del Prete, Z.; Rizzuto, E. Battery Sizing and Composition in Energy Storage Systems for Domestic Renewable Energy Applications: A Systematic Review. *Energies* **2025**, *18*, 5536.
12. Gholami, M.; Muyeen, S.; Mousavi, S.A. Optimal sizing of battery energy storage systems and reliability analysis under diverse regulatory frameworks in microgrids. *Energy Strategy Rev.* **2024**, *51*, 101255.
13. Grimaldi, A.; Minuto, F.D.; Perol, A.; Casagrande, S.; Lanzini, A. Techno-economic optimization of utility-scale battery storage integration with a wind farm for wholesale energy arbitrage considering wind curtailment and battery degradation. *J. Energy Storage* **2025**, *112*, 115500.
14. Pigem, J.M.; García-Muñoz, F.; Bravo, N.J.; Aránguiz, R.; Burgos, V.C. Optimal sizing of community photovoltaic and battery energy storage systems with second-life batteries in peer-to-peer energy communities. *J. Energy Storage* **2026**, *148*, 120031.
15. Levin, T.; Blaisdell-Pijuan, P.L.; Kwon, J.; Mann, W.N. High temporal resolution generation expansion planning for the clean energy transition. *Renew. Sustain. Energy Transit.* **2024**, *5*, 100072.
16. Anderson, O.; Yu, N.; Oikonomou, K.; Maloney, P.; Wu, D. Representative period selection for robust capacity expansion planning in low-carbon grids. In Proceedings of the 2024 IEEE/PES Transmission and Distribution Conference and Exposition (T&D), Anaheim, CA, USA, 6–9 May 2024; IEEE: New York, NY, USA, 2024; pp. 1–5.
17. Parolin, F.; Colbertaldo, P.; Macdonald, R. Improved formulation for long-duration storage in capacity expansion models using representative periods. *arXiv* **2024**, arXiv:2409.19079.
18. Prakash, K.; Ali, M.; Siddique, M.N.I.; Chand, A.A.; Kumar, N.M.; Dong, D.; Pota, H.R. A review of battery energy storage systems for ancillary services in distribution grids: Current status, challenges and future directions. *Front. Energy Res.* **2022**, *10*, 971704.
19. Maeyaert, L.; Vandeveld, L.; Döring, T. Battery storage for ancillary services in smart distribution grids. *J. Energy Storage* **2020**, *30*, 101524.
20. Garttan, G.; Alahakoon, S.; Emami, K.; Jayasinghe, S.G. Battery energy storage systems: Energy market review, challenges, and opportunities in frequency control ancillary services. *Energies* **2025**, *18*, 4174.
21. Ferrucci, T.; Fioriti, D.; Poli, D.; Barberis, S.; Roncallo, F.; Gambino, V. Battery energy storage systems for ancillary services in renewable energy communities. *Appl. Therm. Eng.* **2025**, *260*, 124988.

22. Varhegyi, G.; Nour, M. Integrating fast frequency response ancillary services: A global review of technical, procurement, and market integration challenges. *Clean Energy* **2025**, *9*, 204–218.
23. Fernández-Muñoz, D.; Pérez-Díaz, J.I.; Guisández, I.; Chazarra, M.; Fernández-Espina, Á. Fast frequency control ancillary services: An international review. *Renew. Sustain. Energy Rev.* **2020**, *120*, 109662.
24. Khalilisenobari, R.; Wu, M. Optimal participation of price-maker battery energy storage systems in energy and ancillary services markets considering degradation cost. *Int. J. Electr. Power Energy Syst.* **2022**, *138*, 107924.
25. Alavijeh, N.M.; Khezri, R.; Mazidi, M.; Steen, D.; Tuan, L.A. Optimal scheduling of battery storage systems in the Swedish Multi-FCR market incorporating battery degradation and technical requirements. *arXiv* **2024**, arXiv:2406.07301.
26. Hameed, Z.; Træholt, C. BESS Participation Planning for Provision of Grid Services in Energy and Regulation Markets. In Proceedings of the 2025 IEEE PES 35th Australasian Universities Power Engineering Conference (AUPEC), St. Lucia Campus, Australia, 29 September – 1 October 2025; IEEE: New York, NY, USA, 2025; pp. 1–5.
27. Lauinger, D.; Coté, L.; Sun, A. Storage Participation in Electricity Markets: Arbitrage and Ancillary Services. *arXiv* **2025**, arXiv:2510.10856.
28. Fernández-Muñoz, D.; Pérez-Díaz, J.I. Optimisation models for the day-ahead energy and reserve scheduling of a hybrid wind-battery virtual power plant. *J. Energy Storage* **2023**, *57*, 106296.
29. Superchi, F.; Moustakis, A.; Pechlivanoglou, G.; Bianchini, A. On the importance of degradation modeling for the robust design of hybrid energy systems including renewables and storage. *Appl. Energy* **2025**, *377*, 124645.
30. Wang, J.; Lyu, C.; Bai, Y.; Yang, K.; Song, Z.; Meng, J. Optimal scheduling strategy for hybrid energy storage systems of battery and flywheel combined multi-stress battery degradation model. *J. Energy Storage* **2024**, *99*, 113208.
31. Vykhodtsev, A.V.; Jang, D.; Wang, Q.; Zareipour, H.; Rosehart, W.D. A review of lithium-ion battery models in techno-economic analyses of power systems. *arXiv* **2021**, arXiv:2106.08702.
32. Zhao, C.; Li, X. Microgrid optimal energy scheduling considering neural network based battery degradation. *IEEE Trans. Power Syst.* **2023**, *39*, 1594–1606.
33. Pamshetti, V.B.; Zhang, W.; Andy, M.F.N.; Yan, Q.; Tan, K.T. Enhanced Battery Degradation-Aware Scheduling for Distribution Network with Electric Vehicle Load. In Proceedings of the TENCON 2024-2024 IEEE Region 10 Conference (TENCON), Singapore, 1–4 December 2024; IEEE: New York, NY, USA, 2024; pp. 1044–1047.
34. Mao, Y.; Cai, Z.; Jiao, X.; Long, D. Multi-timescale optimization scheduling of integrated energy systems oriented towards generalized energy storage services. *Sci. Rep.* **2025**, *15*, 8549. <https://doi.org/10.1038/s41598-025-92601-9>.
35. Mao, Y.; Jiao, X. Flexible batch loads as “virtual batteries”: Distributionally robust optimization for computation-electricity co-scheduling in data center integrated energy system. *Electr. Eng.* **2026**, *108*, 246. <https://doi.org/10.1007/s00202-026-03600-6>.
36. Commission Regulation (EU). Commission Regulation (EU) 2017/1485 of 2 August 2017 Establishing a Guideline on Electricity Transmission System Operation. *Off. J. Eur. Union* **2017**, *L 220*, 1–111.
37. De Vos, K.; Stevens, N.; Devolder, O.; Papavasiliou, A.; Hebb, B.; Matthys-Donnadieu, J. Dynamic Dimensioning Approach for Operating Reserves: Proof of Concept in Belgium. *Energy Policy* **2019**, *124*, 272–285. <https://doi.org/10.1016/j.enpol.2018.09.031>.
38. Costilla-Enriquez, N.; Ortega-Vazquez, M.A.; Tuohy, A.; Motley, A.; Webb, R. Operating Dynamic Reserve Dimensioning Using Probabilistic Forecasts. *IEEE Trans. Power Syst.* **2023**, *38*, 603–616. <https://doi.org/10.1109/TPWRS.2022.3163106>.
39. Commission for Electricity and Gas Regulation. Study on the Functioning and Price Evolution of the Belgian Wholesale Electricity Market—Monitoring Report 2023. Technical Report Study (F)2747, Commission for Electricity and Gas Regulation (CREG), Brussels, Belgium, 2024. Available online: <https://www.creg.be/nl/publicaties/studie-f2747> (accessed on 5 June 2026).
40. Elia Transmission Belgium. Design Note on aFRR Balancing Service. Technical report, Elia Transmission Belgium, 2025. Available online: https://www.elia.be/-/media/project/elia/elia-site/electricity-market-and-system/system-services/keeping-the-balance/afrr/202509_design_note_afrr.pdf (accessed on 5 June 2026).
41. Guo, Z.; Wei, W.; Chen, L.; Xie, R.; Mei, S. Sizing Energy Storage to Reduce Renewable Power Curtailment Considering Network Power Flows: A Distributionally Robust Optimisation Approach. *IET Renew. Power Gener.* **2020**, *14*, 3273–3280. <https://doi.org/10.1049/iet-rpg.2020.0354>.
42. Mehrdash, M.; Hobbs, B.F.; Ela, E. Reserve and Energy Scarcity Pricing in United States Power Markets: A Comparative Review of Principles and Practices. *Renew. Sustain. Energy Rev.* **2023**, *183*, 113465. <https://doi.org/10.1016/j.rser.2023.113465>.

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